

Experimental Evaluation of a Financial Education Program in Elementary and Middle School Grades*

Abstract Caio Piza[†] Isabela Furtado[‡] Vivian Amorim[§]

This paper investigates whether providing financial education in elementary and middle school grades improves students' financial proficiency and actual behavior. We use a cluster randomized control trial to evaluate a pilot program implemented in 101 Brazilian municipal schools in 2015. The findings show positive impacts on financial proficiency, mainly among middle school students, and suggestive evidence of improvements in short-term behavioral outcomes. However, the analysis indicates that the program did not impact students' school achievements in both the short and longer terms, which suggests that the program's effects were not strong enough to shift students' behavior decisions.

JEL Codes: D14, I21, I25, O12, O15

1 Introduction

Financial literacy is at the center of the education debate in various developed and developing countries. The Organization for Economic Cooperation and Development (OECD) advocates that financial education should be integrated into basic school curricula (Lusardi and Mitchell, 2014).¹ As individuals are being increasingly exposed to financial decisions at a young age, the sooner they learn to incorporate the costs of their choices in their decision-making process the better. Good financial habits at an early age are likely to benefit schooling, employment, and the standard of living through adulthood (Bruhn et al. (2016); Bruhn et al. (2022)). In fact, developed and developing countries worldwide have been implementing financial education interventions in schools aiming to develop financial skills that could be conducive to better decision-making throughout individuals' life-cycle (Bover et al. (2018), Frisanchi (2020), Gill and Bhattacharya (2019), Luhrmann et al. (2018)), Bruhn et al. (2016), Bhattacharya et al. (2016), De Beckker et al. (2021), Berry et al. (2018), Batty et al. (2020)).

Financial education may improve the quality of intertemporal decision-making by increasing individuals' knowledge about saving and borrowing instruments (Luhrmann et al. (2018)) as

* Acknowledgements: We would like to thank Cláudia Forte, Alzira Silva, Thiago Nascimento, and Cláudia Donega from the Brazilian Association of Financial Education (AEF-Brasil), José Alexandre Vasco from the Securities and Exchange Commission of Brazil (CVM), and the staff from Municipal Secretariats of Education from Manaus and Joinville. We thank the seminar participants from the Sao Paulo School of Economics at Getulio Vargas Foundation, Insper Institute of Education and Research, the 39th grade Brazilian Econometrics Society Meeting (SBE), and the 3ie-IFPRI Seminar Series. We also thank Arianna Legovini, Bilal Zia, Miriam Bruhn, Florence Kondylis, and an anonymous referee for their comments and suggestions. Finally, we thank DIME i2i, DEC Research Support Budget, and the Brazil Country Office for funding support. The reproducibility package is available on the GitHub repository [financial-education-brazil](https://github.com/financial-education-brazil).

[†]Development Impact Evaluation Unit (DIME) of the World Bank. Contact: caiopiza@worldbank.org

[‡]Insper - Institute of Education and Research. Contact: isabelabf@insper.edu.br

[§]Development Impact Evaluation Unit (DIME) of the World Bank. Contact: vamorim@worldbank.org

¹The OECD defines financial literacy as a combination of awareness, knowledge, skills, attitudes, and behaviors necessary to make sound financial decisions and ultimately achieve individual financial well-being. In this study, we use financial education and financial literacy interchangeably.

subjects with higher financial knowledge are more likely to make patient inter-temporal choices (Oberaach and Kaiser (2022)). Also, financial literacy can be a factor in avoiding financial risk and taking advantage of economic opportunities (Miller et al. (2015)). In a recent meta-analysis, Kaiser et al. (2022) find that financial education interventions have, on average, a significant impact on financial knowledge (0.15-0.2 of a standard deviation) and on financial behaviors, such as borrowing, saving and investing, budgeting and planning, insurance, and remittances (0.06–0.1 sd).²

Effective financial education programs that impact children's socio-emotional skills such as patience and self-control could potentially improve outcomes other than the financial-related ones. The more impatient children and adolescents are less likely to invest in human capital and seem more likely to consume alcohol, develop obesity, get pregnant during adolescence, and commit a crime (Castillo et al. 2011; Moffitt et al. 2011; Sutter et al. 2013; Golsteyn et al. 2014; Kautz et al. (2014)). These same socioemotional skills can influence labor market prospects in future life by affecting participation in the labor market, the probability of holding a job, school decisions, and a range of educational outcomes (Acosta et al. (2018); Santos et al. (2021); Alan et al. (2019); Kautz et al. (2014)). From a theoretical standpoint, programs aimed at developing personality traits as such should be prioritized in childhood in the face of the expected higher returns per dollar invested (Heckman et al. 2006; Cunha and Heckman 2007).

There is increasing evidence that financial education programs targeting high school students are effective (Bruhn et al. (2016), Gill and Bhattacharya (2019); Frisanco (2022)). The available literature also suggests that in-school programs not only improve students' financial knowledge but also change their behavior (Bover et al. (2018), Luhrmann et al. (2018), Frisanco (2022), Jamison et al. (2014), Bruhn et al. (2016), Stoddard and Urban (2020), Urban et al. (2020), and Harvey (2019)). Recent evidence suggests that these programs can have long-lasting effects on students' human capital accumulation and on financial decisions and entrepreneurship (Frisanco (2020); Frisanco (2022); Bruhn et al. (2022)).

There is also growing interest and evidence on the positive impacts of financial literacy interventions on elementary and middle school grades. Bhattacharya et al. (2016) find evidence of a 12 percentage points increase in the financial knowledge of eighth-graders in the United States. De Beckker et al. (2021) suggest a 0.46 sd increase in the financial knowledge as well as better financial behavior of eighth and ninth-graders in an experiment in Belgium. Batty et al. (2020) finds evidence of an increase in the financial knowledge of fourth-graders equivalent to 0.25 of a standard deviation in a pilot run in 20 schools in the United States. In an RCT with Turkish elementary schools, Alan and Ertac (2018) find that the program persistently reduces the incidence of behavioral problems in schools and increases students' patience. Berry et al. (2018), on the other hand, do not find impacts of a financial literacy intervention implemented in Ghanaian elementary and junior high schools.

²The literature review includes 76 randomized experiments.

In this paper, we use a cluster randomized controlled trial to investigate the impacts of a financial literacy pilot implemented in 2015 in elementary and middle school grades in two Brazilian municipalities (Manaus and Joinville). We randomly assigned 101 municipal schools to receive the program, and 100 to control. The intervention aimed to help young students become more conscious of the trade-offs embedded in their daily decisions and more forward-looking. The evaluation focused on grades three, five, seven, and nine for reasons discussed later.

We find positive effects on students' financial knowledge of 0.07 of a standard deviation, on average. The effects are driven by middle school grades (0.1 SD) and are consistent with a recent summary of the financial education literature ([Kaiser et al. \(2022\)](#)). We also find suggestive evidence that the program helped students become more conscious about their consumption habits - i.e., more patient and risk-averse. We find suggestive evidence of changes in some contemporary behavioral outcomes such as the use of piggy banks, suggesting some linkages between actual change in behavior in the short-term. However, we do not detect impacts on both contemporary learning outcomes and human capital accumulation. We use a mediation causal analysis to investigate the potential causal link between gains in financial knowledge and changes in behavioral outcomes. Our results do not point to a compelling causal linkage between formal knowledge and change in behavior, a result consistent with [De Becker et al. \(2021\)](#).

Our paper has four main contributions. First, we document the positive impacts on financial proficiency, attitudes, and behavioral outcomes of a large-scale pilot intervention implemented in elementary and middle school grades. Second, we show that financial education programs can potentially impact different skills according to students' life cycles. Third, we test the overlooked hypothesis that increasing financial proficiency is a necessary condition to change individuals' behavior. Finally, we use student-level data from the Brazilian national standardized exams to assess the program's impacts on students' contemporary and longer-term human capital.

Apart from this introduction, this paper is organized as follows. Section [2](#) describes the pilot program. Section [3](#) describes the data collection. Section [4](#) describes the program's implementation. Section [5](#) introduces the empirical strategy, whereas section [6](#) discusses the results. We then conclude.

2 The Pilot

The National Strategy for Financial Education (ENEF) was established by the Brazilian Federal Government in 2010 to promote financial education and empower individuals to make autonomous financial decisions.³ In 2015, the Brazilian Association of Financial Education (AEF-Brasil),⁴ ENEF's implementing partner, developed a financial educational pilot targeting elementary and middle school students, for which the municipalities of Joinville and Manaus volunteered to participate. The intervention was delivered throughout the 2015 school year in a sample of public elementary and middle schools (grades 1–9) managed by the respective municipal governments. The initiative followed an earlier financial literacy program implemented under the same strategy in 892 public high schools across six Brazilian states, which yielded significant improvements in students' financial knowledge, intention to save, and financial autonomy, as well as greater participation in their household finances (Bruhn et al., 2016).

The two pilot municipalities have distinct socioeconomic contexts. Manaus, the capital of Amazonas, is in the less affluent North, while Joinville, in Santa Catarina, is in the relatively affluent South. At the time of the pilot, Joinville exhibited higher values on development indicators, including GDP per capita (USD 23,500 vs. USD 16,300, in 2015 PPP terms), the Municipal Human Development Index (0.809 vs. 0.737), and school attendance among 6- to 14-year-olds (97% vs. 94%).⁵ Together, the two settings span two of the major socioeconomic profiles found in the Brazilian education system, broadening the external validity of our findings within the country while not exhausting its full diversity.

2.1 Teaching Materials

ENEF's teaching materials comprise one student book and one teacher book for each of grades 1–9, of which the pilot evaluates a subset, described below.⁶ These materials follow the OECD's definition of financial education as the process by which individuals improve their understanding of financial concepts and risks and develop the skills and confidence to make informed financial decisions (OECD, 2005). This process operates through three means: information (knowledge

³ENEF stands for *Estratégia Nacional de Educação Financeira* and was created by a Federal Decree – 7.397 of December 22, 2010. It brings together the Ministry of Education, the Central Bank of Brazil, the Securities and Exchange Commission of Brazil (CVM), the Ministry of Finance, the Superintendence of Private Insurance (SUSEP), and the Superintendence of Complementary Pension (PREVIC).

⁴AEF-Brasil stands for *Associação de Educação Financeira do Brasil*.

⁵GDP per capita from the 2015 Municipal Accounts of the Brazilian Institute of Geography and Statistics (IBGE): R\$ 47,233 for Joinville and R\$ 32,598 for Manaus, converted to international dollars using the 2015 PPP conversion factor for Brazil of R\$ 2.01 per international dollar (World Bank). Municipal Human Development Index and school attendance rates from the 2010 Demographic Census, the latest available at the time of the pilot, disclosed by the IBGE.

⁶Materials are publicly available on the CVM's investor education portal at <https://gmw.investidor.gov.br/ensino-fundamental/>. They were published in 2014 by the National Committee of Financial Education (CONEF) and validated by an inter-institutional pedagogical panel.

of concepts), instruction (attitudes and skills), and objective advice on specific financial products, of which the materials cover the first two, with greater emphasis on instruction across all grades.

The content is organized along two dimensions: a spatial dimension that moves from the individual to local, regional, national, and global levels, emphasizing how individual choices reverberate at larger scales; and a temporal dimension that connects past, present, and future decisions, placing the intertemporal nature of financial choice at the center. These dimensions are operationalized through four pedagogical modules, each targeting a grade range: a project-based approach for grades 1–4, a choose-your-own-adventure narrative for grades 5–6, a role-playing simulation game for grades 7–8, and a news-magazine format for grade 9.⁷ Table 1 shows examples for the four pilot-grade books.

The content was not designed to be delivered as a stand-alone subject. Instead, it was designed to be integrated across subjects of the existing curriculum and to be taught by teachers of any specialty, drawing on what the teacher books call the teacher’s role as a citizen rather than as a subject specialist. This integration is built into the material itself: the margins of the teacher books contain icons flagging, for example, connections to specific curricular areas (Portuguese, Mathematics, Environmental Education); the introduction of formal financial concepts (e.g., compound interest, household budget); or psychological biases (e.g., loss aversion, immediacy). We discuss in section 4 how this cross-curricular integration played out in the classroom.

Because each grade range receives a distinct combination of content and pedagogical format, our estimates capture the joint effect of these elements. We discuss grade-specific results with this combined nature in mind in section 6.

2.2 Sampling and Randomization

We defined the sampling frame using the 2015 School Census. In Joinville, we included all 72 elementary and middle schools managed by the municipal government. In Manaus, we sampled 129 of the 302 elementary and middle schools using a two-stage procedure.⁸ **NAO ENTENDI O PROCEDIMENTO COM O IDEB DESCRITO NO FOOTNOTE.** To account for differences in infrastructure and socioeconomic context across the two municipalities, we stratified the

⁷Each pilot-grade teacher book describes its module’s pedagogical approach in Part I, Section 4. The curriculum has 10 learning competencies: debating rights and duties (C01); participating in socially and environmentally responsible financial decisions (C02); distinguishing wants from needs of consumption and saving within the family’s life project (C03); reading and interpreting simple texts in the financial education domain (C04); critically reading advertising texts (C05); participating in financial decisions considering real needs (C06); acting as a multiplier (C07); elaborating a basic financial plan with assistance (C08); and caring for oneself, nature, and common goods, considering both the immediate (C09) and future (C10) consequences of present actions.

⁸We first excluded 53 schools located in riverside communities, following the guidelines of the Department of Education, due to access constraints. Of the remaining 302 schools offering at least one target grade, we sampled 36 of the 202 schools that only offered elementary grades based on the 2013 IDEB quartiles of their grade 5 performance; 28 of the 35 schools that only offered middle school grades based on the 2013 IDEB quartiles of their grade 9 performance; and all 65 schools that offered grades 1–9.

Table 1: Financial education materials by pedagogical module

Module (grade range)	Pedagogical format	Central concepts and attitudes
Project-based (grades 1–4)	Four projects organized around concrete objects of everyday life: a ball (Production and Consumption), the household’s expenses (Organization), the school (Care), and Book Day festivities (Planning).	Price composition; environmentally responsible consumption; needs vs. wants; income and expenses; care for shared public and private spaces; short-term planning and estimation.
Choose-your-own-adventure (grades 5–6)	Three choose-your-own-adventure stories plus a closing group project, centered on environmental sustainability and the “five R’s” (rethink, refuse, reduce, reuse, recycle).	Waste; cash vs. credit purchases; value vs. price; conscious consumption; short-term planning; risk; patrimony; sustainability. Psychological traps introduced: ostentation, immediacy, selective perception, over-confidence.
Role-playing (grades 7–8)	A six-session role-playing game in which student groups represent different economic agents, negotiate proposals, and manage a running budget; new rules are added at each session.	Personal budget (income – expenses = balance); investment and the risk-return trade-off; financing and interest; the financial system; patrimony protection; borrowing. Additional traps: framing, sunk costs, loss aversion.
News-magazine (grade 9)	A print magazine modeled on the visual style of a news website, with reports, interviews, columns, a three-episode web-series, a forum, and hands-on exercises.	Household budgeting; expense categorization; consumer traps; social entrepreneurship; the Consumer Defense Code; public space, public budget, and taxes. Attitudes emphasized: autonomy, risk perception, discipline, and spending control.

Notes: The table summarizes the four pedagogical modules of the program (covering grades 1–9). Each row describes one module based on the corresponding pilot-grade book we examined (grades 3, 5, 7, or 9).

Source: Student and teacher books for grades 1–9, available at <https://gmw.investidor.gov.br/ensino-fundamental/>.

randomization by municipality and by school type: schools offering only elementary grades (grades 1–5), schools offering only middle school grades (grades 6–9), and schools offering all elementary and middle school grades (grades 1–9), generating six strata in total (Table A.2). Within each stratum, we randomly assigned half of the schools to treatment and half to control, resulting in 101 treated and 100 control schools.

Budgetary constraints shaped several aspects of the design. First, the pilot could not include all elementary and middle grades; we selected grades 3, 5, 7, and 9 to generate evidence on each of the four pedagogical modules. Second, we did not conduct a baseline student survey and instead relied on rich school-level administrative data to perform pre-treatment balance tests (section 3). Third, within each school, the program was not delivered to every class of the pilot grades: at the beginning of the 2015 school year, each school principal, in both treatment and control schools, selected a single class at each pilot grade to be included in the survey sample. Across the 201 pilot schools, the median number of classes per pilot grade was 3,⁹ so principals typically chose one of about three classes per grade.

This within-school selection of classes is a potential source of bias, although random assignment was preserved at the school level. Because randomization preceded principals’ class selection,

⁹2015 School Census.

control-school principals had no reason to select classes on characteristics that would correlate with the (counterfactual) intervention. As we show in [section 3](#), observable socioeconomic characteristics, teacher characteristics, and pre-treatment performance are balanced between treatment and control schools, with one exception: students in grade 9 in control schools display higher pre-treatment reading scores. If higher reading correlates with stronger financial proficiency, this imbalance would bias our estimates downward. We discuss the robustness of our estimates to this concern in [section 6](#).

3 Data and Measurement

To evaluate the impact of the financial education pilot, we use a combination of survey and administrative data to measure a broad set of outcomes over different time horizons. Survey data collected at the end of the 2015 school year capture short-term effects on students' financial proficiency, hypothetical attitudes toward consumption and saving, and self-reported behaviors. The survey data were collected right after the program implementation finished, in line with other financial education programs targeting elementary and middle school students (De Beckker et al. (2021), Alan and Ertac (2018), Berry et al. (2018)). Administrative records provide medium- and long-term outcomes, including grade progression, standardized test scores, and dropout rates, observed one to three years after the intervention.

3.1 Survey Data

The Center for Public Policy and Education Evaluation (CAEd), a Brazilian survey firm specializing in standardized assessments and data collection, was responsible for implementing the survey, with students and teachers participating in the pilot. Data collection occurred during the final school days of the 2015 academic year, when CAEd distributed printed questionnaires directly in classrooms. Despite being conducted at the end of the school year—a period when some students may already be on vacation or have dropped out—student participation rates remained high, averaging approximately 80%, comparable to regular school-day attendance. Furthermore, as shown in Table A.5, there is no significant difference between participation rates of treatment and control groups across all grades.

Students completed three paper-based questionnaires during a two-hour session: (i) a financial literacy test, (ii) a questionnaire on hypothetical consumption and savings attitudes and self-reported behaviors, and (iii) a socioeconomic questionnaire. The content of each instrument was tailored to the students' grade level and cognitive development.

The financial literacy test evaluates students' understanding of concepts introduced in the financial education textbooks through multiple-choice questions aligned with grade-appropriate skills (Table A.1). The test includes items that assess students' ability to interpret everyday financial information, such as utility bills or spending records, and to identify the purpose of documents like receipts, checks, and invoices. It also covers savings, expenses, consumption, waste, risk, return, financial planning, and investment. Additional questions assess students' ability to estimate values for simple financial projects, distinguish between paid and unpaid work, interpret financial graphs and tables, and extract relevant financial information from media texts. Not all skills are assessed in all grades; the test content is adapted to the cognitive level of each group, with some skills appearing only in earlier or later grades.¹⁰

¹⁰For third-grade students, the financial literacy test was read aloud by trained enumerators to ensure accessibility,

CAEd used student responses to construct a financial proficiency index using Item Response Theory (IRT), which accounts for the difficulty of each question and generates a standardized score comparable across grades. This measure allows us to assess the impact of the intervention in terms of standard deviations, enabling consistent comparisons within and across student groups.

The second questionnaire assesses students' financial attitudes and self-reported behaviors related to saving and consumption. Attitudinal items present hypothetical scenarios (e.g., "I plan before spending," "I see no problem in owing money"), with responses captured using a four-point Likert scale ("Totally agree" to "Totally disagree"), where 4 reflects the most forward-looking or prudent financial attitude (Table A.4). Based on these responses, CAEd constructed two standardized indices: a consumption index, reflecting students' tendency toward planned versus impulsive spending (self-control), and a savings index, reflecting attitudes related to risk aversion and the importance of saving for the future. These indices are derived using principal component analysis and follow the methodology of psychology-based indices used in similar studies (e.g., Bruhn et al. (2016)). Importantly, these measures reflect students' attitudes, not their underlying preferences or actual financial behavior. This second questionnaire also collects self-reported behavioral outcomes, such as whether students have a piggy bank, receive an allowance, use a credit card, or talk about money with friends or family. These variables are analyzed separately as self-reported behaviors.

The last questionnaire asks about students' socioeconomic background, such as the educational attainment of their mothers. Whether they benefit from the national conditional cash transfer program, *Bolsa Família*. Fifth, seventh, and ninth graders answered all the questionnaires in school. Third-graders answer the financial knowledge questionnaire in school with the help of an enumerator, who reads the questions. Since third-graders are still acquiring reading and writing abilities, this strategy is intended to make the test easier for all students, regardless of their reading and writing skills. Still, for third-graders, the socioeconomic questionnaire and one of the students' attitudes towards savings and consumption were sent to their homes so their legal guardians could fill them out and return them the following day. Legal guardians were supposed to answer it based on how he/she believed his/her child would behave in a given situation. Schools prepared the legal guardians to answer these questionnaires by sending them a note before the test day. Table A.5 shows that over three-quarters of the parents/guardians returned the questionnaire, and that there is no statistically significant difference between the percentage of guardians that completed the questionnaire for treated and control students (WE DID NOT CONSIDER SAVINGS AND CONSUMPTION FOR THIRD GRADERS).

As expected, Table A.7 shows no significant differences between the socioeconomic characteristics of treated and control students. The teachers' questionnaire investigates their wage level, teaching experience, and issues related to the incorporation of financial literacy textbooks in given that students at this age are still developing reading and writing skills.

the curriculum, among other questions. Overall, there are no significant differences between teachers in treatment and control groups (Table A.6).

Overall, our analysis has eight dependent variables from the survey data collected in the field. Three standardized indices have been developed by CAEd, which are (i) financial proficiency, (ii) consumption (a proxy for patience and self-control), and (iii) savings (a proxy for risk aversion). The other five are behavioral ones such as (iv) a dummy indicating whether a student talks to parents about financial subjects, (v) a dummy indicating whether a student talks to friends about the same subject, (vi) students' use piggy banks (a proxy for saving), (vii) students' access to financial services, and (viii) a dummy indicating whether a student receives an allowance from their parents.

3.2 Administrative Data

Due to the limited budget and the richness of publicly available administrative data, we did not collect baseline survey data. Prior to the program's implementation, we used administrative data at the school level to check for a balance between treatment and control schools.

Since 1995, all private and public K-12 schools have participated in the annual School Census. The Census is implemented by the National Institute of Educational Studies and Research (INEP), a research agency under the Brazilian Ministry of Education. The Census collects information on school facilities, school infrastructure, social services, teachers, and students.¹¹ In 2005, INEP introduced the national standardized exam known as *Prova Brasil* that applies proficiency assessments in reading and math to all public schools.¹² Since 1998, INEP has annually applied the National High School Proficiency Test (ENEM).¹³

We merged the pilot data with the surveys above, all of which are public at the student level. Table A.10 shows that treated and control schools do not have significant differences in average students' proficiency in reading and math, grade promotion, dropout, retention rates, and age-grade distortion.¹⁴ However, as presented in section 2, principals' choice of which classes would

¹¹The data is collected at the student level, allowing us to track their sex, ethnicity, age, grade level, instruction time per day, class size, and student outcomes, such as grade promotion, retention, and dropout status.

¹²Schools with at least 20 students enrolled in each evaluated grade. Students take the test at the end of the school year (between October and November). The exam is applied every two years to fifth and ninth-graders, and students in the last high school grade.

¹³ENEM stands for *Exame Nacional do Ensino Médio*. The exam aims to assess students' learning levels in reading, math, human sciences, and natural sciences. ENEM is not mandatory, and the students are responsible for registering for the exam. ENEM participants are the students enrolled in high school, mostly in the last grade, and those who have already finished this level of education. Since 2009, the exam has been used by several Brazilian universities as one of their criteria for selecting their students. Consequently, the number of participants significantly increased, from 115,000 in 1998 to 8.5 million in 2015.

¹⁴Unidentified microdata at the student level is also publicly available and identified data only upon researcher request to access INEP's facilities. We could not request access to students' identified microdata by the beginning of 2015 when the pilot was implemented, and then check the balance between treated and control students. First, we did not have the list of pilot students; second, the request usually takes months to be processed, and it was very

receive the intervention or serve as a control is not necessarily random. In this sense, one may wonder whether school principals from treatment and control schools face different incentives to select the classes they did. In treated schools, principals could either select low-performing classes, hoping to help their students with the intervention, or high-performing classes if they believe these students would benefit most from the program. In control schools, principals could select low performers to show that their schools should also receive the treatment in the future, or high performer classes if they intend to show that their schools do not need the program. In this sense, using administrative data at the school level to check the pre-intervention balance of students' performance has one caveat, since not all the students in these schools are included in the pilot.

To enrich our analysis and compare students instead of schools, in 2019, we requested INEP to merge our pilot data with their administrative datasets at the student level level.¹⁵ Table A.3 shows the variables that we were able to collect and we use them for two main tasks. First, the merged dataset allows us to investigate the impact of the financial literacy pilot on student outcomes only available in School Census, *Prova Brasil*, and *ENEM* such as proficiency in reading and math, grade-promotion, retention and dropout (Table A.3).¹⁶

Second, for seventh and ninth graders included in the pilot, we compare their pre-intervention performance in reading and math, since these students did the proficiency exam when enrolled in fifth grade.¹⁸ Table A.9 shows that, for seventh-grade classes, there are no significant differences between treatment and control classes in pre-treatment students' performance. Nonetheless, for ninth-grade classes, the data show that students in control schools had a significantly higher pre-treatment performance in reading, which might indicate that the principals from control schools selected high-performing classes to be in the pilot evaluation. If this were the case, and if higher reading scores are correlated with better financial proficiency skills, our results would be biased downwards.¹⁹

restrictive then.

¹⁵This task has to be performed by the agency since researchers cannot access identified administrative data at the student level. The INEP technical team performs the merge using students' names and the schools they are enrolled in. Table A.8 and Table A.9 show no significant differences in the percentage of treated and control students found by the INEP technical team in its administrative datasets. All the analyses with administrative data at the student level are run in the cold room of INEP facilities.

¹⁶Table A.3 shows the administrative data that we use to assess the effects of the financial literacy pilot on (i) retention and dropout rates in the year of the pilot or up to three years later (between 2015-2018); on (ii) standardized proficiency test in math and Portuguese in the year of the pilot (for those enrolled in fifth and ninth grades in 2015) and two years after (those enrolled in third and seventh grades in 2015) - for those who did not repeat between third and fifth grades and between seven and ninth grades, respectively; and (iii) ENEM scores and high school completion rates for those enrolled in the ninth grade in the year of the pilot.¹⁷

¹⁸For seventh graders who did not repeat any grade between the fifth and seventh grades, their fifth-grade proficiency is available in 2013. For ninth-graders who did not repeat any grade between the fifth and ninth grades, their fifth-grade proficiency is available in 2011.

¹⁹No standardized proficiency test is applied by INEP before the fifth grade. Therefore, for third and fifth graders, we do not have any administrative data on their pre-treatment performance in reading and math on standardized national exams. Hence, we could not compare students' proficiency between treatment and control classes for these grades.

4 Program Implementation

ENEF provided training on how to incorporate financial literacy textbooks into the school curriculum of the 2015 school year (that started in February and ended at the beginning of December). In Joinville, the training was held in February 2015 for the teaching supervisors of each school. In Manaus, supervisors of the regional education boards were trained at the end of March of the same year. Teachers in the schools that implemented the intervention were trained by those who attended the ENEF training. The pilot schools had the autonomy to decide when to introduce the textbooks into their classes.

The programs' supervisors designed by the Departments of Education of Joinville and Manaus were in charge of monitoring the pilot implementation. During the 2015 school year, the supervisors filled out questionnaires aimed to identify the major bottlenecks and keep track of the use of financial literacy textbooks. ENEF used the findings of the first two questionnaires applied during the first semester of the school year to improve the program's implementation. At the beginning of the second semester, ENEF organized a meeting with teachers in charge of incorporating the pilot textbooks into their classes. This meeting sought to stimulate the use of the textbooks and to encourage teachers who were already using them to share their experiences with those who had had little contact with the material at that point.

Insights into the programs' implementation can also be obtained from the questionnaires applied to teachers and students at the end of the 2015 school year. [Table A.5](#) shows that around 80% of pilot students answered the financial literacy assessment. With the exception of third graders, the socioeconomic questionnaire was answered by at least 97% of those who did the financial literacy test. We observe that there are no significant differences in response rates of treatment and control groups.

[Table A.5](#) shows that almost half of the third-grade teachers introduced financial literacy textbooks during the whole school year. However, middle school teachers only introduced the textbooks in the second semester. Also, according to questionnaires applied to students, more than half of middle-school students received financial literacy textbooks in the second semester. Slightly more than 20% of middle-school students had at least 80% of the financial literacy content covered. The late introduction of the textbooks is reflected in the percentage of the curricula actually covered. One can infer from the data on the program's implementation that the treatment was not very intense, and its intensity differed across schools and grades.

5 Identification Strategy

To estimate the causal impacts of the program on the outcomes of interest, we estimate the following reduced-form regression:

$$Y_{iksm} = \alpha + \delta T_{iksm} + \sum_{n=1}^6 \gamma_n + \epsilon_{iksm} \quad (1)$$

in which Y is the outcome variable (e.g., financial proficiency) of student i in grade k at school s in municipality m , T is a binary variable that is equal to 1 if the student belongs to a school assigned to the program (treatment) and 0 otherwise, γ_n are strata fixed effects, and ϵ is the idiosyncratic error.²⁰

The parameter of interest, δ , measures the intention-to-treat (ITT), the treatment effect on students of schools randomly assigned to receive the treatment. The standard error is clustered at the school level, the randomization unit. It is worth noticing that this regression does not distinguish the effect between grades. To assess the program's effect only for elementary, only for middle school, and for each grade included in the pilot, we separately re-estimate [Equation 1](#) for each one of these samples.²¹ In addition to that, we follow [Firpo et al. \(2009\)](#) and estimate unconditional quantile intention-to-treat (UQITT) effects to check whether the program has distinct impacts across the financial proficiency distribution.²²

²⁰[Table A.2](#) shows that the cluster-randomized trial has six clusters.

²¹We also estimate standard error clustering at the class-school level. The results are very similar and are available upon request.

²²We estimate the unconditional UQITT in two steps. We first run a regression of Y on constant and strata dummies and save the residuals. We then use the residuals as a dependent variable to estimate the UQITT.

6 Results

6.1 Financial Proficiency, Savings and Consumption Attitudes, and Behavioral Outcomes

Table 2 shows that the intervention has a positive and statistically significant impact of 0.07 of a standard deviation on students' financial proficiency, with the effect driven by the middle school grades. Larger impacts on students of more advanced grades are consistent with the hypothesis that financial education programs are more effective in increasing the financial proficiency of the students more likely to experience financial decisions in daily life (Frisancho (2020), Bruhn et al. (2016)), and Zia (2023). Our results are in line with Jamison et al. (2014) as the authors find an impact of 0.08 of a standard deviation of a financial literacy pilot implemented among Ugandan youth groups, but are considerably smaller when compared to the ones reported by De Beckker et al. (2021) and Batty et al. (2020) who find an increase in the financial literacy of middle and elementary grade students of 0.46 and 0.25 of a standard deviation, respectively.

Our estimates also point to positive impacts on consumption and saving indices, the ones measuring students' attitudes, particularly among fifth-graders (Table 2).²³ These estimates should be interpreted as children becoming more conscious, self-controlled, and forward-looking. Interestingly, the magnitude of the impact on saving attitudes declines steadily with school grades. This pattern suggests that older students are less risk-averse and potentially self-controlled.

The larger point estimates for self-control and risk aversion among fifth graders might be associated with a few factors. Younger students might be less overconfident than teenagers and this can make them more responsive to interventions aimed to develop risk-aversion behavior. At the same token, it might be easier to change the behavior of younger students since their financial habits are not consolidated yet.

To investigate whether the intervention had distributional impacts on financial proficiency, we estimate the quantile treatment effects.²⁴ Figure 1 displays the effect of the pilot on elementary school students at 20 different quantiles of this distribution. The UQITT estimates show a positive and statistically significant of about 0.05 of an SD starting in the 45 percentile of the financial proficiency distribution. The results suggest that the program benefited a subset of elementary school students, but the effect on the subset was almost fully counterbalanced by no effects for the bottom 40 percentile. For middle-school students, the UQITT and ITT estimates

²³Due to characteristics of the research instruments for the third graders, the variables employed to set up the savings and consumption index are unavailable for this grade. Thus, column (2) estimates correspond to only fifth-graders.

²⁴Since the estimates are on the eligible population and do not control for covariates, we call it unconditional quantile intent to treat effects (UQITT)

Table 2: ITT estimates for financial proficiency and hypothetical attitudes towards savings and consumption

	(1) Pooled	(2) Elementary	(3) Middle school	(4) 3 rd grade	(5) 5 th grade	(6) 7 th grade	(7) 9 th grade
Financial Proficiency							
Treatment	0.073 (0.034)	0.047 (0.046)	0.099 (0.042)	0.053 (0.077)	0.043 (0.046)	0.090 (0.048)	0.106 (0.055)
pvalue	0.033	0.307	0.022	0.495	0.35	0.061	0.057
pvalue RI	0.034	0.328	0.024	0.502	0.381	0.055	0.064
N. obs	14655	7641	7014	3635	4006	3640	3374
Number of clusters	201	172	145	170	168	144	144
R2	0.167	0.15	0.188	0.127	0.173	0.208	0.168
Strata F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Consumption index self-control							
Treatment	0.094 (0.026)	0.122 (0.042)	0.082 (0.031)		0.122 (0.042)	0.080 (0.041)	0.084 (0.037)
pvalue	0	0.004	0.01		0.004	0.054	0.025
pvalue RI	0.002	0.006	0.009		0.009	0.052	0.029
N. obs	10220	3648	6572		3648	3409	3163
Number of clusters	200	167	145		167	144	142
R2	0.089	0.119	0.076		0.119	0.1	0.055
Strata F.E.	Yes	Yes	Yes	-	Yes	Yes	Yes
Saving index risk-aversion							
Treatment	0.055 (0.025)	0.087 (0.041)	0.044 (0.029)		0.087 (0.041)	0.066 (0.040)	0.018 (0.038)
pvalue	0.031	0.036	0.138		0.036	0.102	0.644
pvalue RI	0.027	0.042	0.114		0.042	0.105	0.645
N. obs	10022	3549	6473		3549	3368	3105
Number of clusters	200	166	145		166	144	142
R2	0.046	0.096	0.025		0.096	0.032	0.019
Strata F.E.	Yes	Yes	Yes	-	Yes	Yes	Yes

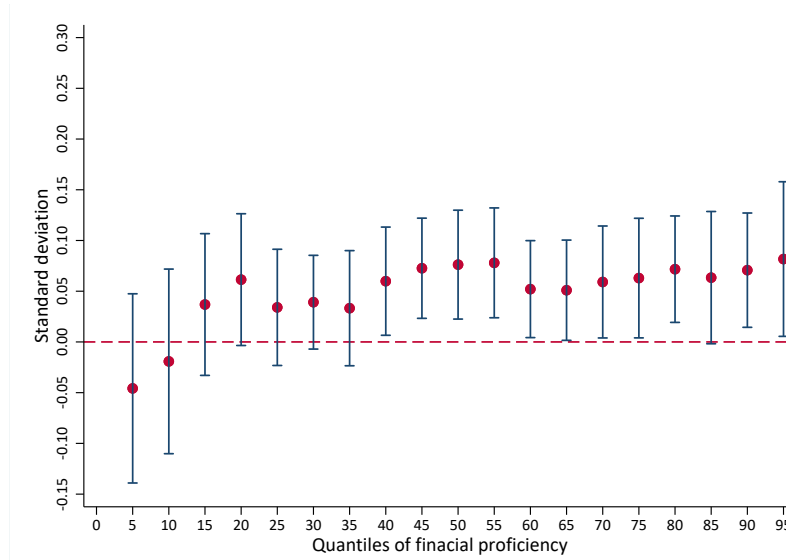
Note: Clustered standard errors at the school level in parenthesis. RI stands for Randomization Inference. The financial proficiency is computed by CAEd based on students' answers on a standardized exam that uses the Item Response Theory (IRT). The consumption and saving indices are based on students' choices in questions containing four possible answers (I totally agree, I agree, I disagree, I totally disagree) and they are aimed to measure students' self-control and risk-aversion, respectively. The financial proficiency, savings, and consumption indices are normalized so that the treatment effect can be interpreted in terms of standard deviations (SD).

are similar, pointing to a more homogeneous effect across the distribution in more advanced grades (see Figure B.2 in the appendix).

We further explore whether the program affected students' behavioral outcomes by looking at more objective measures of change in behavior such as the use of piggy banks, and investments in human capital. We expect that more patient students would be more likely to invest in human capital whose returns happen later in life.

Table 3 shows impacts on students' likelihood to discuss financial subjects with their parents and friends, their likelihood to use a piggy bank (a proxy for actual savings), and their likelihood to receive a monthly allowance from parents (a proxy for students' access to informal sources of credit). Finally, we test whether the treatment increases children's access to financial services,

Figure 1: Unconditional Quantile ITT estimates for financial proficiency, elementary students - 95%CI



Source: The financial proficiency is computed by CAEd based on student's answers on a standardized exam that used the Item Response Theory (IRT). The index is normalized so that the treatment effect can be measured in terms of standard deviations (SD). We first run this financial proficiency score on the treatment dummy, the six strata dummies, and on a constant. We then take the residual of this estimation and run a simultaneous quantile regression, for each one of the quantiles shown in the Figure, in which the dependent variable is the residual of the first regression and the independent variable is the treatment dummy. 1000 bootstrap replications.

such as debit and credit cards.²⁵ ITT estimates show that fifth graders are 5.7 pp (or 11.6 percent over the control mean) and 5.6 pp (or 28 percent over the control mean) more likely to talk to parents and friends about financial subjects respectively, and 2.7 pp. (or 71 percent over the control mean) more likely to use piggy banks (a proxy for actual savings). Overall, these results point to positive effects on awareness and saving outcomes. For middle school students, we find some indication that they become more likely to talk to friends about financial subjects. Unlike Bruhn et al. (2016), we do not find any impact of the program on students' access to financial services. This is unsurprising in our context given that most students are young and do not deal with financial transactions in their daily lives.

Our estimates show that the program had larger impacts on financial proficiency in later grades and larger effects on attitudes and behavior in earlier grades. The results then suggest that the association between financial proficiency and behavioral change might not be straightforward. Individuals might have formal knowledge about different subjects, such as financial proficiency, but the application of the knowledge in real life can be discouraged by the institutional environment they are inserted (Carpena and Zia (2018)). To formally test the hypothesis that learning mediates change in behavior, we follow Imai et al. (2011) and decompose the treatment effect estimates into the average causal mediation effect (ACME), and the average causal direct effect

²⁵To measure students' access to financial services, we use a dummy variable that is one if a student has access to at least one service and zero otherwise.

(ADE).²⁶ The goal of this exercise is to assess the share of the change in behavioral outcomes that could be attributed to improvements in financial proficiency (ACME) and the share that comes directly from the program (ADE). The decomposition of ITT effects on ACME and ADE relies on the sequential ignorability assumption. The first part of the assumption assumes that conditioned on pre-treatment confounders the treatment assignment is orthogonal to the potential outcomes and the mediator outcome (financial proficiency). The second part of the assumption implies that the ‘observed mediator is ignorable given the actual treatment status and pre-treatment confounders (Imai et al., 2011). This second part of the assumption is the strongest, even in the presence of random assignment. It poses that conditional on the actual treatment (not treatment assignment) and a vector of pre-treatment confounders, the outcome is independent of the mediator factor. Note that, in our case, this implies an assumption that attitudinal and behavioral outcomes are orthogonal to financial proficiency given the actual treatment and once we control for a vector of school characteristics. Since we did not design the experiment to obtain clean estimates of the ACME and ADE, we interpret our results as suggestive evidence.

Table 3: ITT estimates for behavioral outcomes

	Pooled	Elementary students	Middle students		Pooled	Elementary students	Middle students
Talk to parents				Talk to friends			
Treatment	0.012 (0.010)	0.057 (0.019)	-0.012 (0.012)	Treatment	0.035 (0.009)	0.056 (0.014)	0.023 (0.011)
pvalue	0.216	0.003	0.306	pvalue	0	0	0.044
pvalue of RI	0.222	0.004	0.345	pvalue of RI	0	0	0.058
Obs.	10748	3875	6873	Obs.	10725	3864	6861
Number of clusters	200	167	145	Number of clusters	200	167	145
R2	0.006	0.003	0.001	R2	0.008	0.012	0.005
Strata F.E.	Yes	Yes	Yes	Strata F.E.	Yes	Yes	Yes
Piggy's bank use				Use of financial services			
Treatment	0.022 (0.011)	0.027 (0.015)	0.017 (0.013)	Treatment	0.002 (0.009)	-0.002 (0.012)	0.006 (0.013)
pvalue	0.042	0.067	0.205	pvalue	0.783	0.86	0.662
pvalue of RI	0.046	0.063	0.236	pvalue of RI	0.773	0.872	0.638
Obs.	12739	5917	6822	Obs.	12538	5718	6820
Number of clusters	201	172	145	Number of clusters	201	172	145
R2	0.028	0.038	0.01	R2	0.005	0.004	0.006
Strata F.E.	Yes	Yes	Yes	Strata F.E.	Yes	Yes	Yes
Allowance							
Treatment	0.013 (0.008)	0.017 (0.014)	0.01 (0.010)				
pvalue	0.12	0.222	0.303				
pvalue of RI	0.111	0.245	0.305				
Obs.	10699	3862	6837				
Number of clusters	200	167	145				
R2	0.009	0.012	0.007				
Strata F.E.	Yes	Yes	Yes				

Note: Clustered standard errors at the school level. RI stands for Randomization Inference. The dependent variables are equal to 1 if students talk to parents or friends about financial subjects, if they use a piggy bank, financial services, or receive an allowance from their parents, and 0 otherwise.

²⁶See the online appendix for a formal discussion of Casual Mediation Effects.

According to the estimates in [Table A.13](#), while we find some evidence that financial proficiency mediates the improvement of middle students' hypothetical attitudes in scenarios involving savings and consumption, it does not mediate changes in their behavior. For elementary school students, all of the effects seem to be direct, i.e., not mediated by changes in financial proficiency ([Table A.13](#)). In other words, the ADE and ACME estimates suggest that financial education programs affect students' attitudes and behavior through pathways other than the accumulation of formal knowledge. Overall, the estimates do not support the hypothesis that changes in financial proficiency are a necessary condition to change individuals' behavioral outcomes, a finding consistent with [Carpena and Zia \(2018\)](#), and [De Becker et al. \(2021\)](#).

To assess whether the results on both financial proficiency, attitudes, and behavioral outcomes are picking up some heterogeneity in the data, we carried out a series of subgroup analyses. First, we compared the effects on boys and girls. The point estimates are larger for boys, but the difference in estimates between boys and girls is not statistically significant in most cases. Second, we tested whether the impacts varied by poverty status. We aimed to assess whether poorer kids - those participating in the Brazilian conditional cash transfer program - would respond less to an intervention intended to help kids become more forward-looking. In fact, the estimates on financial proficiency and consumption attitudes are larger in absolute terms for the poor, but the differences in coefficients between poor and non-poor are not statistically significant for the most part. The heterogeneous effect analyses are available in the online appendix.

6.2 Robustness Check

Back in 2015, we did not have access to administrative student-level data to check the balance between treatment and control groups and investigate the impacts of the program on students' performance in reading and math. As described in [section 3](#), in 2020 and 2021 INEP granted us access to its facilities to complement our analyses using student-level data. These data allow us to rerun the balance tables at the student level and, more importantly, look at the program's impacts on both contemporary and longer-term human capital outcomes.

As discussed in [section 2](#), school principals' choice of the classrooms to participate in the pilot might not have been random. Consequently, if school principals strategically picked the best classrooms aiming to show the benefits of the program, or the worst classrooms betting on a potential complementarity between financial education and other school subjects, our estimates would be biased as we could be comparing treated and control students with a different pre-treatment level of skills. To assess students' pre-treatment level of proficiency in Portuguese and math, we look at their scores in the Brazilian standardized exam when in grade 5. Indeed, the average test scores in grade 5 for students attending the seventh and ninth grades in 2015 were higher in the control group ([Table A.9](#)). To check whether this imbalance in pre-treatment

skill level affects our results, we re-estimate [Equation 1](#) for seventh and ninth graders controlling for their pre-treatment math and reading test scores.²⁷

Controlling for past performance in math and reading leads to a substantial gain in precision. Among seventh graders, we observe a slight increase in the point estimate for financial proficiency scores (from 0.09 SD to 0.098 SD). For ninth-graders, we find a meaningful increase in financial proficiency scores, with the point estimate jumping from 0.11 SD to 0.15 SD ([Table A.14](#)), though the difference is not statistically significant. The updated estimates indicate that we would underestimate the impacts of the pilot on financial proficiency if we did not control for an imbalance in pre-treatment proficiency in math and reading at the student level, particularly for ninth graders. Controlling for past performance in math and reading scores does not change the point estimates for consumption and saving attitudes.

6.3 Implementation and IV Results

The following analysis seeks to shed some light upon the role played by the quality of the program’s implementation on its impacts. This exercise is useful in that it avoids hasty conclusions that a given intervention did not have the expected effect when in fact the program was not carried out as planned. We use a variable that informs the percentage of the syllabus (use of the teaching material) covered during the school year. As discussed earlier, the material was delivered somewhat late at the treated schools, and that eventually hindered the exposure to the program and consequently impacted the intensity of treatment. Approximately one-third of the teachers claimed they covered up to 60% of the syllabus. We first carry out a correlation analysis testing whether the effects are associated with the quality of the program’s implementation. We do so by splitting the treatment group into two groups: schools with up to 60% of the syllabus covered, and schools with more than 60% of the syllabus covered.

Since the quality of the program’s implementation is endogenous, we follow [Frisancho \(2020\)](#) and use the random assignment to treatment as an instrumental variable for the program’s implementation. The IV estimates provide the local average treatment effect (LATE) of the program by comparing the subgroup of treated schools that received the treatment (supervisor/teacher training and teachers who used the textbooks) with the control schools. We create a binary variable, Z , that is 1 for schools with at least 60% of the syllabus covered during the academic year, and 0 otherwise leveraging information gathered from teachers and students on whether the textbooks were incorporated into the classroom during the school year.²⁸ Since the data show that some students and teachers from control schools apparently received and used the textbooks, we

²⁷Seventh and ninth-graders are the ones for which the pre-treatment performance in reading and math is available. For seventh-graders, we have their fifth-grade performance in reading and math in 2013. For ninth-graders, we have their fifth-grade performance in 2011.

²⁸Unfortunately, the variable is categorical and we are unable to define the dummy at the median of the distribution or to estimate marginal treatment effects.

use a two-stage least squares to estimate the LATE:

$$\text{First-Stage: } Z_{iksm} = \gamma_1 + \gamma_2 T_{hksm} + \Sigma C_j + \omega_{1iksm}, h = 1, 2. \quad (2)$$

$$\text{Second-Stage: } Y_{iksm} = \lambda_1 + \lambda_2 \widehat{Z_{iksm}} + \Sigma C_j + \omega_{2iksm} \quad (3)$$

The parameter of interest, λ_2 , is the LATE. The IV estimates are shown in Table 4. The estimates are slightly larger and less precise than ITT ones, suggesting that the program did not have large effects even when better implemented.

Table 4: IV estimates

	Pooled		Elementary		Middle school	
Financial Proficiency	ITT	LATE	ITT	LATE	ITT	LATE
Estimate	0.073	0.085	0.047	0.055	0.099	0.115
N. obs	(0.034)	(0.039)	(0.046)	(0.054)	(0.042)	(0.049)
R2	14,655	14,655	7,641	7,641	7,014	7,014
	0.167	0.166	0.15	0.149	0.188	0.189
Consumption Index						
Estimate	0.094	0.110	0.122	0.146	0.082	0.095
N. obs	(0.026)	(0.031)	(0.042)	(0.050)	(0.031)	(0.036)
R2	10,220	10,220	3,648	3,648	6,572	6,572
	0.089	0.087	0.119	0.117	0.076	0.076
Savings Index						
Estimate	0.055	0.064	0.087	0.104	0.044	0.051
N. obs	(0.025)	(0.029)	(0.041)	(0.049)	(0.029)	(0.034)
R2	10,022	10,022	3,549	3,549	6,473	6,473
	0.046	0.046	0.096	0.095	0.025	0.025

Note: Clustered standard errors at the school level. Students are considered treated if their teachers answered that at least part of the content of financial education was covered in class. For 8.5% of the treated students and 7.5% of the control students, we do not have a teacher assigned. In these cases, we cannot infer actual treatment using teachers' answers on the usage of the material. We then consider a student as treated if at least 50% of her peers in the classroom state they received the financial literacy textbook.

6.4 Longer-term Effects on Behavioral Outcomes: Investment in Human Capital

In this section, we assess whether the pilot impacts students' human capital investment decisions. The hypothesis is that if students understand the importance of education for their future earnings and become more patient and forward-looking following the pilot, they would be more likely to study harder, potentially improve their academic performance in reading and math, and be less likely to drop out of school (Bruhn et al. (2016)). This has a direct policy implication as it would indicate whether offering financial education to elementary and middle school students could

improve individuals' outcomes other than their financial proficiency with potential benefits to their financial health and labor market outcomes in the longer term.

We start by checking if the program has an immediate impact on the performance of fifth and ninth-graders in math and reading in 2015, the year the financial education program was piloted.²⁹ Figure B.1 shows a significant correlation between students' financial proficiency score and their reading and math performance. We then investigate whether there is any causal relationship in place. Our results do not point to a contemporary effect of financial education on either math or reading skills for both grades (Table 5). Although the nil effects on math and reading proficiency of fifth graders could be explained by the fact that the intervention had a small effect on a subset of the students as suggested by the UQITT estimates, the lack of treatment effects for grade 9 is less expected.

These results seem to suggest that teachers involved in this pilot evaluation did not prioritize financial education at the expense of other school subjects (e.g., teach to the test), a concern the research team and some school principals had back in the day. Overall, our estimates are in line with Batty et al. (2020) that also finds evidence of no substitution from the core school curriculum.

Besides the contemporary effects on learning outcomes, the data allow us to test whether the program impacted students' human capital accumulation. We first assess whether the pilot affected students' probability of being retained or dropping out of school between 2015 and 2018. Second, for third and seventh graders, we assess their performance in math and reading two years after the intervention (2017) when they reached fifth and ninth grades, respectively.³⁰ Finally, for ninth-graders, we assess whether the intervention affects their probability of finishing high school, taking the ENEM test, and the test takers' ENEM proficiency score.

We do not find any indication that the financial education program affects students' academic achievements – test scores and school progression – in the longer term (Table 5). We interpret these findings as evidence that the program does not make students become more forward-looking with regard to their human capital accumulation decisions, which might be associated with poor program implementation as described in section 4.

²⁹Fifth and ninth grades are the ones for which the performance in math and reading is available in 2015 due to their participation in *Prova Brasil*. See Table A.3.

³⁰For those who did not repeat between third and fifth grades and between seventh and ninth graders, respectively.

Table 5: ITT estimates for Human Capital Accumulation Outcomes

	3rd grade		5th grade		7th grade			9th grade		
	Only strata	Strata FE Financial proficiency	Only strata	Strata FE Financial proficiency	Only strata	Strata FE Financial proficiency	Strata FE + Performance 5th grade	Only strata	Strata FE Financial proficiency	Strata FE + Performance 5th grade
Repetition	0.017 (0.017)	0.010 (0.019)	0.029 (0.014)	0.025 (0.015)	-0.007 (0.014)	-0.002 (0.015)	-0.008 (0.016)	0.016 (0.016)	0.031 (0.017)	0.018 (0.018)
Dropout	0.005 (0.006)	0.007 (0.006)	0.011 (0.010)	0.000 (0.008)	0.006 (0.018)	0.002 (0.014)	0.002 (0.012)	0.033 (0.020)	0.028 (0.018)	0.020 (0.019)
Portuguese Prova Brasil	-0.029 (0.047)	-0.053 (0.051)	-0.017 (0.046)	-0.032 (0.031)	-0.004 (0.047)	-0.057 (0.041)	-0.005 (0.034)	-0.015 (0.042)	-0.063 (0.036)	0.028 (0.036)
Math Prova Brasil	-0.066 (0.053)	-0.096 (0.055)	-0.033 (0.049)	-0.045 (0.039)	-0.005 (0.051)	-0.060 (0.048)	-0.009 (0.038)	-0.077 (0.042)	-0.094 (0.037)	-0.020 (0.039)
Finish high school								-0.035 (0.025)	-0.035 (0.024)	-0.019 (0.023)
Participation in ENEM								-0.025 (0.024)	-0.016 (0.022)	-0.012 (0.023)
Portuguese ENEM								-0.046 (0.036)	-0.004 (0.05)	-0.025 (0.031)
Math ENEM								-0.028 (0.047)	-0.035 (0.063)	0.002 (0.045)
Obs.	4650	3635	4755	4006	4697	3640	3137	4455	3374	2851

Note: Clustered standard errors at the school level.

Grade retention assumes a value of 1 if the student is retained in the same grade at the end of the school year, and 0 otherwise. Dropout is equal to 1 if the student drops out at the end of the school year, and 0 otherwise. Finish high school is equal to 1 if the student finishes high school, and 0 otherwise. Participating in ENEM is equal to 1 if the students participate in the ENEM assessment, and 0 otherwise. Portuguese and Math are the proficiency scores in *Prova Brasil* on the SAEB scale.

7 Conclusion

In this paper, we use a cluster randomized controlled trial to evaluate the impacts of a financial literacy pilot program in Brazilian elementary schools during the academic year of 2015. The pilot was implemented in 101 municipal schools and involved approximately 9,000 students. Our main findings showed that the program improved students' financial proficiency, consumption (self-control), and saving (risk-aversion) attitudes, and some behavioral outcomes, such as the use of piggy banks. We find strong indications of heterogeneous effects though. While proficiency gains were stronger among middle school students, changes in attitudes and behavioral outcomes stood out among elementary education students.

We tested the overlooked hypothesis in this literature that poses that change in formal knowledge precedes changes in actual behavior. We use a causal mediation effect analysis to test this hypothesis. Overall, our results seem to reject the hypothesis that claims that changes in knowledge are a necessary condition to change individuals' habits.

Finally, we used student test score data in math and reading in different school grades to investigate the program's impacts on contemporary learning outcomes and on human capital accumulation. We do not find evidence that the program impacted grade progression, retention, dropout rates, or learning outcomes. Our reading of the results is that the program did not affect students' intertemporal decision-making, confirming the preliminary results used by the implementing partner who decided to scale the program down in its current format.

References

- ACOSTA, P. M. ET AL. (2018): “The role of cognitive and socio-emotional skills in labor markets,” *IZA World of Labor*.
- ALAN, S., T. BONEVA, AND S. ERTAC (2019): “Ever failed, try again, succeed better: Results from a randomized educational intervention on grit,” *The Quarterly Journal of Economics*, 134, 1121–1162.
- ALAN, S. AND S. ERTAC (2018): “Fostering patience in the classroom: Results from randomized educational intervention,” *Journal of Political Economy*, 126, 1865–1911.
- BATTY, M., J. M. COLLINS, C. O’ROURKE, AND E. ODDERS-WHITE (2020): “Experiential financial education: A field study of my classroom economy in elementary schools,” *Economics of Education Review*, 78, 102014.
- BERRY, J., D. KARLAN, AND M. PRADHAN (2018): “The impact of financial education for youth in Ghana,” *World Development*, 102, 71–89.
- BHATTACHARYA, R., A. GILL, AND D. STANLEY (2016): “The effectiveness of financial literacy instruction: The role of individual development accounts participation and the intensity of instruction,” *Journal of Financial Counseling and Planning*, 27, 20–35.
- BOVER, O., L. HOSPIDO, AND E. VILLANUEVA (2018): “The impact of high school financial education on financial knowledge and choices: Evidence from a randomized trial in Spain,” .
- BRUHN, M., G. GARBER, S. KOYAMA, AND B. ZIA (2022): “The Long-Term Impact of High School Financial Education,” .
- BRUHN, M., L. D. S. LEÃO, A. LEGOVINI, R. MARCHETTI, AND B. ZIA (2016): “The Impact of High School Financial Education: Evidence from a Large-Scale Evaluation in Brazil,” *American Economic Journal: Applied Economics*, 8, 256–295.
- CARPENA, F. AND B. ZIA (2018): *The Causal Mechanism of Financial Education: Evidence from Mediation Analysis*, World Bank Policy Research Working Paper no. 8619.
- CASTILLO, M., P. J. FERRARO, J. L. JORDAN, AND R. PETRIE (2011): “The today and tomorrow of kids: Time preferences and educational outcomes of children,” *Journal of Public Economics*, 95, 1377–1385.
- CUNHA, F. AND J. HECKMAN (2007): “The technology of skill formation,” *American Economic Review*, 97, 31–47.

- DE BECKKER, K., K. DE WITTE, AND G. VAN CAMPENHOUT (2021): “The effect of financial education on students’ consumer choices: Evidence from a randomized experiment,” *Journal of Economic Behavior & Organization*, 188, 962–976.
- FIRPO, S., N. M. FORTIN, AND T. LEMIEUX (2009): “Unconditional quantile regressions,” *Econometrica*, 77, 953–973.
- FRISANCHO, V. (2020): “The impact of financial education for youth,” *Economics of Education Review*, 78, 101918.
- (2022): “Is School-Based Financial Education Effective? Immediate and Long-Lasting Impacts on High School Students,” *The Economic Journal*.
- GILL, A. AND R. BHATTACHARYA (2019): “The effects of a financial literacy intervention on the financial and economic knowledge of high school students,” *The Journal of Economic Education*, 50, 215–229.
- GOLSTEYN, B. H., H. GRÖNQVIST, AND L. LINDAHL (2014): “Adolescent time preferences predict lifetime outcomes,” *The Economic Journal*, 124, F739–F761.
- HARVEY, M. (2019): “Impact of financial education mandates on younger consumers’ use of alternative financial services,” *Journal of Consumer Affairs*, 53, 731–769.
- HECKMAN, J. J., J. STIXRUD, AND S. URZUA (2006): “The effects of cognitive and noncognitive abilities on labor market outcomes and social behavior,” *Journal of Labor economics*, 24, 411–482.
- IMAI, K., L. KEELE, D. TINGLEY, AND T. YAMAMOTO (2011): “Unpacking the black box of causality: Learning about causal mechanisms from experimental and observational studies,” *American Political Science Review*, 105, 765–789.
- JAMISON, J. C., D. KARLAN, AND J. ZINMAN (2014): “Financial education and access to savings accounts: Complements or substitutes? Evidence from Ugandan youth clubs,” Tech. rep., National Bureau of Economic Research.
- KAISER, T., A. LUSARDI, L. MENKHOFF, AND C. URBAN (2022): “Financial education affects financial knowledge and downstream behaviors,” *Journal of Financial Economics*, 145, 255–272.
- KAUTZ, T., J. J. HECKMAN, R. DIRIS, B. TER WEEL, AND L. BORGHANS (2014): “Fostering and measuring skills: Improving cognitive and non-cognitive skills to promote lifetime success,” .

- LUHRMANN, M., M. SERRA-GARCIA, AND J. WINTER (2018): “The Impact of Financial Education on Adolescents’ Intertemporal Choices,” *American Economic Journal: Economic Policy*, 10, 309–32.
- LUSARDI, A. AND O. S. MITCHELL (2014): “The Economic Importance of Financial Literacy: Theory and Evidence,” *Journal of Economic Literature*, 52, 5–44.
- MILLER, M., J. REICHELSTEIN, C. SALAS, AND B. ZIA (2015): “Can You Help Someone Become Financially Capable? A Meta-Analysis of the Literature,” *The World Bank Research Observer*, 30, 220–246.
- MOFFITT, T. E., L. ARSENEAULT, D. BELSKY, N. DICKSON, R. J. HANCOX, H. HARRINGTON, R. HOUTS, R. POULTON, B. W. ROBERTS, S. ROSS, ET AL. (2011): “A gradient of childhood self-control predicts health, wealth, and public safety,” *Proceedings of the National Academy of Sciences*, 108, 2693–2698.
- OBERRAUCH, L. AND T. KAISER (2022): “Cognitive ability, financial literacy, and narrow bracketing in time-preference elicitation,” *Journal of Behavioral and Experimental Economics*, 98, 101844.
- OECD (2005): *Improving Financial Literacy: Analysis of Issues and Policies*, Paris: OECD Publishing.
- SANTOS, I., V. PETROSKA-BESKA, P. CARNEIRO, L. ESKREIS-WINKLER, A. M. MUNOZ BOUDET, I. BERNIELL, C. KREKEL, O. ARIAS, AND A. DUCKWORTH (2021): “Can Grit Be Taught? Lessons from a Nationwide Field Experiment with Middle-School Students,” .
- STODDARD, C. AND C. URBAN (2020): “The effects of state-mandated financial education on college financing behaviors,” *Journal of Money, Credit and Banking*, 52, 747–776.
- SUTTER, M., M. G. KOCHER, D. GLÄTZLE-RÜTZLER, AND S. T. TRAUTMANN (2013): “Impatience and uncertainty: Experimental decisions predict adolescents’ field behavior,” *American Economic Review*, 103, 510–31.
- URBAN, C., M. SCHMEISER, J. M. COLLINS, AND A. BROWN (2020): “The effects of high school personal financial education policies on financial behavior,” *Economics of Education Review*, 78, 101786.
- ZIA, B. (2023): “When is financial education successful? Taking stock of the new wave of field evidence,” *Handbook of Microfinance, Financial Inclusion and Development*, 119–133.

A Tables

Table A.1: Descriptors and skills on the financial knowledge test

		3 rd grade	5 th grade	7 th grade	9 th grade
D01	Being able to identify the subject of texts whose topic explores socially responsible attitudes towards the environment	yes	yes	yes	yes
D02	Being able to find information in texts about consumption – light, water, telephone bills, among others	yes	yes	no	no
D03	Being able to identify the purpose of texts and text formats that include expenses, consumption, spending	yes	yes	no	no
D04	Being able to recognize the purpose of text genres related to finances – receipts, checks, invoices	no	yes	yes	yes
D05	Being able to recognize situations in which concepts related to finances are present: savings, expenses, consumption, spending, waste, risk, return, financial planning, and investment, among others.	no	no	yes	yes
D06	Being able to identify situations related to financially responsible attitudes	no	yes	yes	yes
D07	Being able to find information in graphs and tables that contain data related to finances (purchases, sales, spending)	yes	no	no	no
D08	Being able to find information in texts that circulate in the financial world: classified ads, news features, among others	yes	yes	yes	yes
D09	Being able to estimate values and/or procedures necessary for financial projects	yes	yes	yes	yes
D10	Being able to distinguish remunerated from non-remunerated work.	yes	yes	no	no
D11	Being able to identify the origin and destination of varied products and/or those that can be recycled.	yes	no	no	no
D12	Being able to recognize socially responsible situations related to public and private spaces.	no	no	no	yes
D13	Being able to identify advantages, disadvantages, and risks of cash and credit sales.	no	yes	no	yes
D14	Being able to find implicit information in media texts that are relevant for decision-making in finances.	no	yes	yes	yes

Table A.2: Sample selected for the pilot study

Schools offering:	(A) Joinville				(B) Manaus				Total Manaus and Joinville	
	I	II	III		IV	V	VI		VII	
	Number of schools	Randomization Sample	Randomization Groups		Number of schools	Randomization Sample	Randomization Groups		Number of randomized schools	
			Treatment	Control			Treatment	Control	Treatment	Control
1st-5th grades	Stratum A.1 20	All	10	10	Stratum B.1 202	36 sampled based on 2013 IDEB 1st-5th grades	18	18	28	28
6th-9th grades	Stratum A.2 2	All	1	1	Stratum B.2 35		14	14	15	15
1st-9th grades	Stratum A.3 50	All	25	25	Stratum B.3 65	All	33	32	58	57
Total	72		36	36	302	129	65	64	101	100

Notes: Number of schools managed by the municipal governments of Joinville and Manaus (School Census, 2015). Following the guidelines of the Department of Education, 53 schools located in riverside communities in Manaus are not included in the pilot, due to difficulties in accessing some of these areas. In Joinville, all 72 schools are included in the randomization. In Manaus, 36 out of 202 schools offering first to fifth grade are randomized into treatment and control groups. The 36 schools were selected based on the distribution of the Educational Development Index for those grades in 2013. Also, in Manaus, 28 out of 35 schools offering sixth to ninth grade are randomized into treatment and control. The 28 schools were selected based on the distribution of the Educational Development Index of those grades in 2013. Finally, all schools offering first to ninth grade in Manaus are randomized into treatment and control. Overall, six strata are created in this randomization process (two municipalities and within the municipalities three types of schools).

Table A.3: Dependent variables collected in *Prova Brasil*, School Census and ENEM.

Survey Data		Administrative Data	
<i>We matched the pilot students with the administrative data</i>			
Pilot students in 2015	Proficiency in Math and Portuguese (Prova Brasil)	Grade-promotion/dropout (School Census)	ENEM
3rd graders in 2015 →	Their 5th grade proficiency in 2017	If they ever repeated or dropped between 2015-2018	
5th graders in 2015 →	Their 5th grade proficiency in 2015 and their 7th grade proficiency in 2017	If ever repeated or dropped between 2015-2018	
7th graders in 2015 →	Their 5th grade proficiency in 2013 and their 9th grade proficiency in 2017	If ever repeated or dropped between 2015-2018	
9th graders in 2015 →	Their 5th grade proficiency in 2011	If ever repeated or dropped between 2015-2018 and if finished high school	ENEM average score in 2018

Note: Retention rates, dropout, ENEM scores, and high school completion rates are available annually (School Census and ENEM disclosed by INEP). Performance in Portuguese and Math is only available every two years for fifth and ninth-graders (*Prova Brasil* disclosed by INEP).

Table A.4: Examples of questions on students' behavior towards savings and consumption

Statement	Answers' scale			
	Totally agree	Agree	Disagree	Totally disagree
Consumption questions				
I buy what I want, then I see how I can pay	1	2	3	4
I see no problem in owing money	1	2	3	4
If the brand is famous the product is of high quality	1	2	3	4
The best product is always the most expensive	1	2	3	4
I plan before spending my money	4	3	2	1
It is worthless to plan because the money comes from luck	1	2	3	4
Buy what I want is more important than planning	1	2	3	4
Saving questions				
I think that saving money is important to avoid problems in the future	4	3	2	1
I feel safer when I can save some money	4	3	2	1
Saving some money is important to avoid debt	4	3	2	1
Buying everything I want is more important than putting the money together	1	2	3	4
Avoiding waste is also a way to save money	4	3	2	1
I try to use the products for longer	4	3	2	1
Whenever I can, I save money	4	3	2	1
I would rather spend the change on something I want than save the money for later	1	2	3	4

Table A.5: Program implementation, 2015

	3rd grade			5th grade			7th grade			9th grade		
Students' participation rate												
	C	T	T-C	C	T	T-C	C	T	T-C	C	T	T-C
Mean	77.4	79.0	-1.7	83.9	84.6	-0.7	76.7	78.3	-1.6	77.0	74.4	2.6
Standard error	(0.9)	(0.9)		(0.8)	(0.7)		(0.9)	(0.9)		(0.9)	(0.9)	
Number of obs	2380	2270		2375	2380		2376	2321		2222	2233	
Response rate of socioeconomic and attitudinal questionnaires												
	C	T	T-C	C	T	T-C	C	T	T-C	C	T	T-C
Mean	65.3	65.6	-0.3	97.4	99.1	-1.7	99.6	99.7	-0.2	97.7	99.3	-1.6
Standard error	(1.1)	(1.1)		(0.4)	(0.2)		(0.2)	(0.1)		(0.4)	(0.2)	
Number of obs	1808	1756		1992	2014		1822	1818		1712	1638	
Deliver of the financial literacy textbooks according to students, in %												
	Mean	Sd		Mean	Sd		Mean	Sd		Mean	Sd	
Haven't received	-	-		5.69	23.1		9.70	29.6		5.60	23.0	
In the end of the year	-	-		11.1	31.5		15.8	36.5		18.0	38.4	
By the middle of the year	-	-		57.8	49.3		57.6	49.4		53.8	49.8	
Beginning of the year	-	-		25.3	43.4		16.8	37.4		22.4	41.7	
% of classes by the semester in which the teachers used the financial literacy textbooks												
Treated classes												
	Mean	sd		Mean	sd		Mean	sd		Mean	sd	
Only in 1st semester	24.6	43.3		17.7	38.4		17.9	38.6		20.3	40.5	
Only in 2nd semester	27.1	44.7		41.7	49.6		52.2	50.3		57.8	49.7	
All year	48.1	50.2		40.5	49.4		29.8	46.1		21.8	41.6	
% of classes by the percentage of the financial literacy textbook covered by the teachers												
Treated classes												
	Mean	Sd		Mean	Sd		Mean	Sd		Mean	Sd	
Less than 40%	6.09	24.0		0	0		5.97	23.8		7.69	26.8	
Between 40% and 60%	26.8	44.5		21.9	41.6		25.3	43.8		27.6	45.0	
Between 60% and 80%	40.2	49.3		40.2	49.3		44.7	50.1		36.9	48.6	
More than 80%	26.8	44.5		37.8	48.7		23.8	42.9		27.6	45.0	

Notes: ENEF questionnaires were applied to teachers and students at the end of the 2015 school year. T = treatment; C = control. The value displayed for t-tests is the differences in the means across the groups. The covariate variable strata are included in all estimation regressions. The second block (response rate of socioeconomic and attitudinal questionnaires) refers only to students who answered the financial proficiency test.

Table A.6: Balance test for teachers' characteristics

		3rd			5th			7th			9th		
		C	T	T-C	C	T	T-C	C	T	T-C	C	T	T-C
Teachers characteristics, in %													
Gender: male	Mean	7.2	9.9	-2.6	10.0	20.9	-10.9	48.5	33.6	14.8	38.6	44.6	-6.0
	SE	(2.9)	(3.3)		(3.4)	(4.3)		(6.2)	(4.6)		(5.9)	(5.0)	
	Obs	83	81		80	91		66	107		70	101	
Age: less than 35 yers	Mean	27.7	26.8	0.9	26.3	30.4	-4.2	35.8	33.6	2.2	26.8	31.7	-4.9
	SE	(4.9)	(4.9)		(5.0)	(4.8)		(5.9)	(4.6)		(5.3)	(4.7)	
	Obs	83	82		80	92		67	107		71	101	
Age: 36 to 50 yers	Mean	50.6	59.8	-9.2	60.0	53.3	6.7	44.8	54.2	-9.4	53.5	50.5	3.0
	SE	(5.5)	(5.4)		(5.5)	(5.2)		(6.1)	(4.8)		(6.0)	(5.0)	
	Obs	83	82		80	92		67	107		71	101	
Age: older than 51 years	Mean	21.7	13.4	8.3	13.8	16.3	-2.6	17.9	11.2	6.7	19.7	16.8	2.9
	SE	(4.6)	(3.8)		(3.9)	(3.9)		(4.7)	(3.1)		(4.8)	(3.7)	
	Obs	83	82		80	92		67	107		71	101	
Color: white	Mean	45.8	42.7	3.1	48.8	38.0	10.7	43.3	35.5	7.8	46.5	38.6	7.9
	SE	(5.5)	(5.5)		(5.6)	(5.1)		(6.1)	(4.6)		(6.0)	(4.9)	
	Obs	83	82		80	92		67	107		71	101	
Experience: up to 5 years	Mean	19.3	17.1	2.2	11.3	19.6	-8.3	17.9	15.0	3.0	14.1	14.9	-0.8
	SE	(4.4)	(4.2)		(3.6)	(4.2)		(4.7)	(3.5)		(4.2)	(3.6)	
	Obs	83	82		80	92		67	107		71	101	
Experience: 6 to 15 years	Mean	41.0	46.3	-5.4	42.5	41.3	1.2	40.3	57.0	-16.7	52.1	45.5	6.6
	SE	(5.4)	(5.5)		(5.6)	(5.2)		(6.0)	(4.8)		(6.0)	(5.0)	
	Obs	83	82		80	92		67	107		71	101	
Experience: more than 16 years	Mean	39.8	36.6	3.2	45.0	38.0	7.0	41.8	27.1	14.7	33.8	39.6	-5.8
	SE	(5.4)	(5.4)		(5.6)	(5.1)		(6.1)	(4.3)		(5.7)	(4.9)	
	Obs	83	82		80	92		67	107		71	101	
Wage: 3 to 4 minimum wages	Mean	30.1	19.5	10.6	31.3	22.8	8.4	10.4	27.1	-16.7	19.7	21.8	-2.1
	SE	(5.1)	(4.4)		(5.2)	(4.4)		(3.8)	(4.3)		(4.8)	(4.1)	
	Obs	83	82		80	92		67	107		71	101	
Wage: 4 to 5 minimum wages	Mean	31.3	25.6	5.7	33.8	38.0	-4.3	32.8	20.6	12.3	21.1	28.7	-7.6
	SE	(5.1)	(4.8)		(5.3)	(5.1)		(5.8)	(3.9)		(4.9)	(4.5)	
	Obs	83	82		80	92		67	107		71	101	
Wage: 5 to 6 minimum wages	Mean	4.8	17.1	-12.3	16.3	16.3	-0.1	19.4	23.4	-4.0	25.4	29.7	-4.4
	SE	(2.4)	(4.2)		(4.2)	(3.9)		(4.9)	(4.1)		(5.2)	(4.6)	
	Obs	83	82		80	92		67	107		71	101	
Wage: more than 6 minimum wages	Mean	6.0	7.3	-1.3	6.3	9.8	-3.5	16.4	11.2	5.2	15.5	5.9	9.6
	SE	(2.6)	(2.9)		(2.7)	(3.1)		(4.6)	(3.1)		(4.3)	(2.4)	
	Obs	83	82		80	92		67	107		71	101	

Notes: T = treatment; C = control.

We run the balance test controlling for strata.

All the variables shown in the table were collected using the questionnaires CAEd applied to teachers of pilot schools.

Table A.7: Balance test for households and students' characteristics

		3rd			5th			7th			9th		
Household characteristics, in %													
Access to paved street	Mean	68.7	66.8	1.9	67.6	67.4	0.2	71.5	69.4	2.1	73.7	75.1	-1.4
	SE	(1.4)	(1.5)		(1.1)	(1.1)		(1.1)	(1.1)		(1.1)	(1.1)	
	Obs	1087	1028		1914	1983		1804	1802		1662	1616	
Access to electricity	Mean	99.6	99.1	0.5	95.9	96.0	-0.1	97.6	97.6	-0.0	97.9	97.0	0.9
	SE	(0.2)	(0.3)		(0.5)	(0.4)		(0.4)	(0.4)		(0.4)	(0.4)	
	Obs	1041	1005		1914	1980		1800	1803		1668	1621	
Access to piped water	Mean	96.7	96.3	0.4	96.1	96.2	-0.2	96.5	96.7	-0.3	97.3	96.8	0.5
	SE	(0.5)	(0.6)		(0.4)	(0.4)		(0.4)	(0.4)		(0.4)	(0.4)	
	Obs	1060	1023		1925	1988		1806	1808		1667	1618	
Access to garbage collection	Mean	91.5	93.3	-1.8	85.7	84.3	1.4	85.2	88.7	-3.5	90.2	91.2	-1.1
	SE	(0.9)	(0.8)		(0.8)	(0.8)		(0.8)	(0.7)		(0.7)	(0.7)	
	Obs	1050	1002		1907	1961		1794	1796		1668	1609	
Beneficiary of Bolsa Família	Mean	36.2	36.1	0.2	45.4	44.9	0.4	40.9	42.9	-1.9	38.3	35.8	2.6
	SE	(1.5)	(1.5)		(1.1)	(1.1)		(1.2)	(1.2)		(1.2)	(1.2)	
	Obs	1082	1059		1905	1958		1790	1788		1648	1604	
Student characteristics, in %													
Adequate age for their grade	Mean	87.9	86.6	1.3	82.3	80.6	1.7	79.9	80.1	-0.2	81.2	81.1	0.0
	SE	(1.0)	(1.1)		(0.9)	(0.9)		(0.9)	(0.9)		(1.0)	(1.0)	
	Obs	1038	997		1902	1952		1781	1781		1636	1602	
Mother education/ Legal guardian: incomplete elementary school	Mean	26.1	28.2	-2.1	26.3	25.2	1.1	25.1	26.9	-1.8	24.1	22.8	1.3
	SE	(1.3)	(1.4)		(1.2)	(1.2)		(1.2)	(1.2)		(1.2)	(1.1)	
	Obs	1081	1050		1272	1333		1347	1317		1364	1335	
Mother education/ Legal guardian: incomplete high school	Mean	20.9	20.9	0.0	18.1	18.8	-0.7	22.3	18.4	3.9	18.6	20.1	-1.5
	SE	(1.2)	(1.3)		(1.1)	(1.1)		(1.1)	(1.1)		(1.1)	(1.1)	
	Obs	1081	1050		1272	1333		1347	1317		1364	1335	
Mother education/ Legal guardian: at least high school degree	Mean	53.0	51.0	2.1	55.7	56.0	-0.4	52.6	54.7	-2.1	57.3	57.0	0.3
	SE	(1.5)	(1.5)		(1.4)	(1.4)		(1.4)	(1.4)		(1.3)	(1.4)	
	Obs	1081	1050		1272	1333		1347	1317		1364	1335	
Gender: male	Mean				51.3	50.1	1.2	48.0	48.1	-0.1	49.5	49.6	-0.1
	SE				(1.1)	(1.1)		(1.2)	(1.2)		(1.2)	(1.2)	
	Obs				1905	1964		1798	1799		1660	1614	
Color: white	Mean				40.6	37.3	3.2	36.5	34.9	1.6	37.0	35.5	1.5
	SE				(1.1)	(1.1)		(1.1)	(1.1)		(1.2)	(1.2)	
	Obs				1888	1945		1771	1787		1634	1585	

Notes: T = treatment; C = control.

We run the balance test controlling for strata. All the variables shown in the table were collected using the questionnaires CAEd applied to students of pilot schools. For each grade, the number of observations in the treatment and control groups can change depending on the variable under evaluation. This happens when students did not answer that specific question. [Table A.11](#) shows that there are no significant differences between the treatment and control groups regarding the percentage of these variables that are missing. [Table A.12](#) shows the results of the multivariate test of means. Except for seventh grade, we do not reject the hypothesis that the means of students' characteristics are equal in the treatment and control groups.

Table A.8: Pre-treatment balance test for third and fifth graders

	(1) Control		(2) Treatment		T-Test Difference (1)-(2)
	N	Mean/SE	N	Mean/SE	
Third grade					
Beneficiary of Bolsa Família	1082	36.23 (3.10)	1059	36.07 (2.92)	0.16
Adequate age for their grade	1038	86.99 (1.49)	997	85.66 (1.76)	1.34
Gender: male	2071	53.84 (0.92)	2009	51.57 (1.22)	2.27
Color: white	1624	38.92 (3.80)	1582	36.47 (3.94)	2.44
Mother education/Legal guardian: at least high school degree	1081	53.01 (2.59)	1050	50.95 (2.26)	2.05
Student found in School Census 2015-2018	2380	87.02 (1.60)	2270	88.50 (1.51)	-1.49
Student found in Prova Brasil 2017	2380	59.03 (1.84)	2270	59.16 (1.89)	-0.13
Prova Brasil reading score in 5th grade 2017	1405	232.76 (2.51)	1343	230.43 (2.45)	2.33
Prova Brasil math score in 5th grade 2017	1405	243.09 (2.92)	1343	238.55 (3.09)	4.54
Fifth grade					
Beneficiary of Bolsa Família	1905	45.35 (2.80)	1958	44.94 (2.65)	0.41
Adequate age for their grade	1902	78.39 (1.71)	1952	77.61 (1.58)	0.78
Gender: male	2302	51.87 (1.12)	2307	50.59 (1.10)	1.28
Color: white	1888	40.57 (2.52)	1945	37.33 (2.48)	3.25
Mother education/Legal guardian: at least high school degree	1272	55.66 (1.87)	1333	56.04 (2.00)	-0.38
Student found in School Census 2015-2018	2375	89.94 (1.52)	2380	90.04 (1.08)	-0.11
Student found in Prova Brasil 2015	2375	79.33 (2.22)	2380	81.60 (1.77)	-2.27
Prova Brasil reading score in 5th grade 2017	1884	220.58 (2.92)	1942	218.95 (2.68)	1.62
Prova Brasil math score in 5th grade 2017	1884	231.83 (3.82)	1942	229.17 (3.33)	2.66

Notes: We run the balance test controlling for strata. All the variables shown in the table are from *Prova Brasil*, and School Census.

Table A.9: Pre-treatment balance test for seventh and ninth graders

	(1) Control		(2) Treatment		T-Test Difference (1)-(2)
	N	Mean/SE	N	Mean/SE	
Seventh grade					
Beneficiary of Bolsa Família	1790	40.95 (2.92)	1788	42.90 (2.87)	-1.95
Adequate age for their grade	1781	77.20 (2.18)	1781	76.81 (1.85)	0.39
Gender: male	1798	48.00 (1.45)	1799	48.08 (1.04)	-0.08
Color: white	2226	31.63 (4.09)	2186	32.20 (3.89)	-0.58
Mother education/Legal guardian: at least high school degree	1347	52.64 (2.01)	1317	54.75 (2.05)	-2.11
Student found in School Census 2015-2018	2376	91.20 (1.66)	2321	91.12 (1.11)	0.08
Student found in Prova Brasil 2013	2376	67.05 (2.25)	2321	66.52 (1.91)	0.52
Prova Brasil reading score in 5th grade 2013	1593	211.03 (3.16)	1544	209.28 (2.68)	1.75
Prova Brasil math score in 5th grade 2013	1593	226.11 (4.10)	1544	224.70 (3.66)	1.41
Student found in Prova Brasil 2017	2376	62.50 (2.04)	2321	61.83 (1.88)	0.67
Prova Brasil reading score in 9th grade 2017	1485	273.81 (2.58)	1435	274.74 (2.31)	-0.93
Prova Brasil math score in 9th grade 2017	1485	269.66 (3.48)	1435	271.07 (3.01)	-1.42
Ninth grade					
Beneficiary of Bolsa Família	1648	38.35 (3.07)	1604	35.79 (2.59)	2.56
Adequate age for their grade	1636	76.96 (1.80)	1602	76.09 (2.30)	0.86
Gender: male	2161	49.56 (1.05)	2174	47.10 (1.08)	2.46
Color: white	2074	32.64 (4.22)	2087	30.09 (3.95)	2.55
Mother education/Legal guardian: at least high school degree	1364	57.26 (1.75)	1335	57.00 (1.77)	0.25
Student found in School Census 2015-2018	2222	92.26 (0.98)	2233	92.79 (0.84)	-0.53
Student found in Prova Brasil 2011	2222	63.77 (72) (2.24)	2233	64.22 (72) (2.06)	-0.45
Prova Brasil reading score in 5th grade 2011	1417	206.10 (71) (3.07)	1434	200.39 (72) (2.78)	5.71
Prova Brasil math score in 5th grade 2011	1417	221.42 (71) (3.80)	1434	215.72 (72) (3.59)	5.69
Student found in Prova Brasil 2015	2222	82.36 (72) (1.31)	2233	81.10 (72) (1.45)	1.26
Prova Brasil reading score in 9th grade 2015	1830	265.01 (70) (2.27)	1811	264.14 (70) (1.92)	0.87
Prova Brasil math score in 9th grade 2015	1830	264.88 (70) (2.73)	1811	260.73 (70) (2.74)	4.15
Graduate in High School students 9grade	2222	59.63 (72) (1.58)	2233	56.02 (72) (1.91)	3.61
Student in ENEM 2018 students 9grade	2222	46.17 (72) (1.54)	2233	43.62 (72) (1.83)	2.56
Nota padronizada média ENEM-2018	793	0.13 (71) (0.04)	723	0.12 (71) (0.05)	0.01

Notes: We run the balance test controlling for strata. All the variables shown in the table are from *Prova Brasil*, and School Census.

Table A.10: Pre-treatment balance test with administrative data at school level

	Joinville		Manaus	
	Beta	Obs	Beta	Obs
% teachers in 1st to 5th grade with undergrad	-0.00	70	0.86	100
% teachers in 6th to 9th grade with undergrad	-1.10	51	-0.36	91
Age grade distortion 3rd grade (2013)	-2.33	66	-2.06	100
Age grade distortion 5th grade (2013)	-1.86	68	3.73	100
Age grade distortion 7th grade (2013)	-2.70	51	-0.92	90
Age grade distortion 9th grade (2013)	-2.53	50	3.31	89
Grade-promotion 7th grade (2009)	-1.31	45	2.50	70
Grade-promotion 9th grade (2009)	-0.81	45	-2.88	70
Grade-promotion 7th grade (2011)	-0.53	51	-0.69	77
Grade-promotion 9th grade (2011)	-0.47	51	-2.65	76
Grade-promotion 7th grade (2013)	-1.04	51	-0.68	86
Grade-promotion 9th grade (2013)	-0.48	51	0.00	86
Grade-promotion 3rd grade (2009)	1.75	60	1.81	82
Grade-promotion 5th grade (2009)	0.54	60	-0.04	82
Grade-promotion 3rd grade (2011)	0.65	61	-0.02	88
Grade-promotion 5th grade (2011)	0.49	62	-2.25	88
Grade-promotion 3rd grade (2013)	0.37	59	0.26	97
Grade-promotion 5th grade (2013)	0.93	60	-2.83	97
Math performance 9th grade (2009)	-7.53	45	-0.00	69
Reading performance 9th grade (2009)	-2.12	45	1.40	69
Math performance 9th grade (2011)	-3.41	51	0.41	74
Reading performance 9th grade (2011)	-6.52	51	-2.00	74
Math performance 9th grade (2013)	-4.27	51	-2.55	85
Reading performance 9th grade (2013)	-2.87	51	-0.71	85
Math performance 5th grade (2009)	-4.81	59	0.72	79
Reading performance 5th grade (2009)	-2.94	59	-0.04	79
Math performance 5th grade (2011)	-5.59	62	-0.48	86
Reading performance 5th grade (2011)	-3.85	62	1.19	86
Math performance 5th grade (2013)	-3.04	59	0.74	97
Reading performance 5th grade (2013)	-3.22	59	0.43	97
IDEB 9th grade (2009)	-0.18	45	0.06	69
IDEB 9th grade (2011)	-0.20	51	-0.02	74
IDEB 9th grade (2013)	-0.16	51	-0.03	85
IDEB 5th grade (2009)	-0.08	59	0.01	79
IDEB 5th grade (2011)	-0.14	62	0.02	86
IDEB 5th grade (2013)	-0.12	59	-0.03	97

Note: $p < 0.01$, $p < 0.05$, $p < 0.1$ The table shows the coefficients of the dummy variable that identifies whether the selected schools are assigned to the treatment or control group. These coefficients are obtained from regressions of the above-listed covariates on the treatment dummy, the six strata dummies, and a constant. Clustered standard errors at the school level are in parentheses

Table A.11: Regression of the percentage of students' characteristics that are missing on treatment status

	Third-grade		Fifth-grade		Seventh-grade		Ninth-grade	
	Beta	Obs	Beta	Obs	Beta	Obs	Beta	Obs
Access to paved street	0.00	4,650	-0.03	4,755	-0.02	4,697	0.02	4,455
Access to electricity	-0.00	4,650	-0.03	4,755	-0.02	4,697	0.02	4,455
Access to piped water	-0.00	4,650	-0.03	4,755	-0.02	4,697	0.02	4,455
Access to garbage collection	0.00	4,650	-0.03	4,755	-0.02	4,697	0.03	4,455
Beneficiary of Bolsa Família	-0.01	4,650	-0.03	4,755	-0.02	4,697	0.02	4,455
Adequate age for their grade	-0.00	4,650	-0.02	4,755	-0.02	4,697	0.02	4,455
Mother education: incomplete elementary school	-0.01	4,650	-0.03	4,755	0.00	4,697	0.01	4,455
Mother education: incomplete high school	-0.01	4,650	-0.03	4,755	0.00	4,697	0.01	4,455
Mother education: at least high school degree	-0.01	4,650	-0.03	4,755	0.00	4,697	0.01	4,455
Gender: male			-0.03	4,755	-0.02	4,697	0.02	4,455
Color: white			-0.03	4,755	-0.02	4,697	0.02	4,455

Note: $p < 0.01$, $p < 0.05$, $p < 0.1$. The table shows the coefficients of a dummy variable that identifies whether the students are in the treatment or control group. The dependent variables are the percentage of missing observations for each one of the above-mentioned variables. These variables are regressed on the treatment dummy, the six strata dummies, and a constant. Clustered standard errors at the school level are in parentheses.

Table A.12: Multivariate test of means

	Test	Statistic	F	df1	df2	pvalue
Third-grade	Lawley	.006	1.32	8	1694	.224
Third-grade	Pillai	.006	1.32	8	1694	.224
Third-grade	Roy	.006	1.32	8	1694	.224
Third-grade	Wilks	.993	1.32	8	1694	.224
Fifth-grade	Lawley	.004	1.04	10	2330	.401
Fifth-grade	Pillai	.004	1.04	10	2330	.401
Fifth-grade	Roy	.004	1.04	10	2330	.401
Fifth-grade	Wilks	.995	1.04	10	2330	.401
Seventh-grade	Lawley	.007	1.84	10	2452	.047
Seventh-grade	Pillai	.007	1.84	10	2452	.047
Seventh-grade	Roy	.007	1.84	10	2452	.047
Seventh-grade	Wilks	.992	1.84	10	2452	.047
Ninth-grade	Lawley	.005	1.33	10	2479	.208
Ninth-grade	Pillai	.005	1.33	10	2479	.208
Ninth-grade	Roy	.005	1.33	10	2479	.208
Ninth-grade	Wilks	.994	1.33	10	2479	.208

Note: The table shows the results of multiple-sample multivariate tests on means. We test whether the means of several students' characteristics are the same between the treatment and control groups. For third-graders, we test the variables: lives on one paved street, access to electricity, access to piped water, a beneficiary of Bolsa Família, age-grade distortion, and mothers' education. For fifth, seventh, and ninth graders, we also include students' sex and the color of their skin.

Table A.13: Average causal mediation effects

POOLED																					
	Consumption			Saving			Talk to parents			Talk to friends			Piggy's bank use			Use of financial services			Allowance		
	Mean	95% CI		Mean	95% CI		Mean	95% CI		Mean	95% CI		Mean	95% CI		Mean	95% CI		Mean	95% CI	
Total effect	.033	.018	.049	.023	.011	.034	.001	.000	.002	-.00	-.00	-.00	.001	.000	.002	-.00	-.00	-.00	-.00	-.00	.000
ADE	.058	.023	.094	.030	-.00	.068	.014	-.00	.034	.038	.020	.055	.016	-.00	.033	.010	-.00	.024	.015	-.00	.031
ACME	.092	.053	.130	.053	.012	.093	.016	-.00	.035	.036	.019	.054	.017	.000	.035	.007	-.00	.021	.014	-.00	.031
% of total	.366	.260	.635	.424	.244	1.61	.093	-.46	.756	-.03	-.06	-.02	.094	-.04	.542	-.26	-3.6	3.13	-.02	-.17	.081
effect mediated																					
Primary education students																					
	Consumption			Saving			Talk to parents			Talk to friends			Piggy's bank use			Use of financial services			Allowance		
	Mean	95% CI		Mean	95% CI		Mean	95% CI		Mean	95% CI		Mean	95% CI		Mean	95% CI		Mean	95% CI	
Total effect	.020	-.00	.046	.012	-.00	.034	-.00	-.00	.000	-.00	-.00	.001	-.00	-.00	.000	.001	-.00	.003	-.00	-.00	.000
ADE	.102	.040	.165	.077	.011	.145	.063	.031	.096	.058	.031	.086	.009	-.01	.036	.021	-.00	.046	.003	-.01	.023
ACME	.122	.056	.188	.089	.021	.158	.063	.031	.096	.056	.030	.083	.009	-.01	.036	.023	-.00	.048	.001	-.02	.022
% of total	.163	.105	.351	.139	.078	.529	-.00	-.00	-.00	-.03	-.07	-.02	-.00	-.03	.029	.063	-.26	.464	-.13	-4.0	2.11
effect mediated																					
Middle school students																					
	Consumption			Saving			Talk to parents			Talk to friends			Piggy's bank use			Use of financial services			Allowance		
	Mean	95% CI		Mean	95% CI		Mean	95% CI		Mean	95% CI		Mean	95% CI		Mean	95% CI		Mean	95% CI	
Total effect	.040	.019	.061	.027	.012	.041	.003	.001	.005	.002	.000	.003	-.00	-.00	-.00	-.00	-.00	.000	.000	-.00	.001
ADE	.037	-.00	.083	.009	-.03	.059	-.01	-.03	.012	.012	-.01	.039	.016	-.00	.038	.023	.002	.045	.024	.002	.048
ACME	.077	.031	.124	.036	-.01	.084	-.00	-.03	.015	.014	-.00	.041	.013	-.00	.035	.022	.001	.045	.025	.003	.048
% of total	.529	.326	1.27	.687	-4.8	5.92	-.20	-2.8	2.82	.114	-.88	1.37	-.15	-1.9	1.31	-.02	-.15	-.01	.015	.007	.108
effect mediated																					

Note: The dependent variables are equal to 1 if students talk to parents or friends about financial subjects, if they use a piggy bank, financial services, or receive an allowance from their parents, and 0 otherwise. Sensitive analysis runs using the command *medeff* in *Stata*. Standard errors are estimated with bootstrap with 1000 repetitions. ACME: Average Casual Mediation Effect. ADE: Average Direct Effect.

Table A.14: ITT estimates on Proficiency and Attitudinal Outcomes

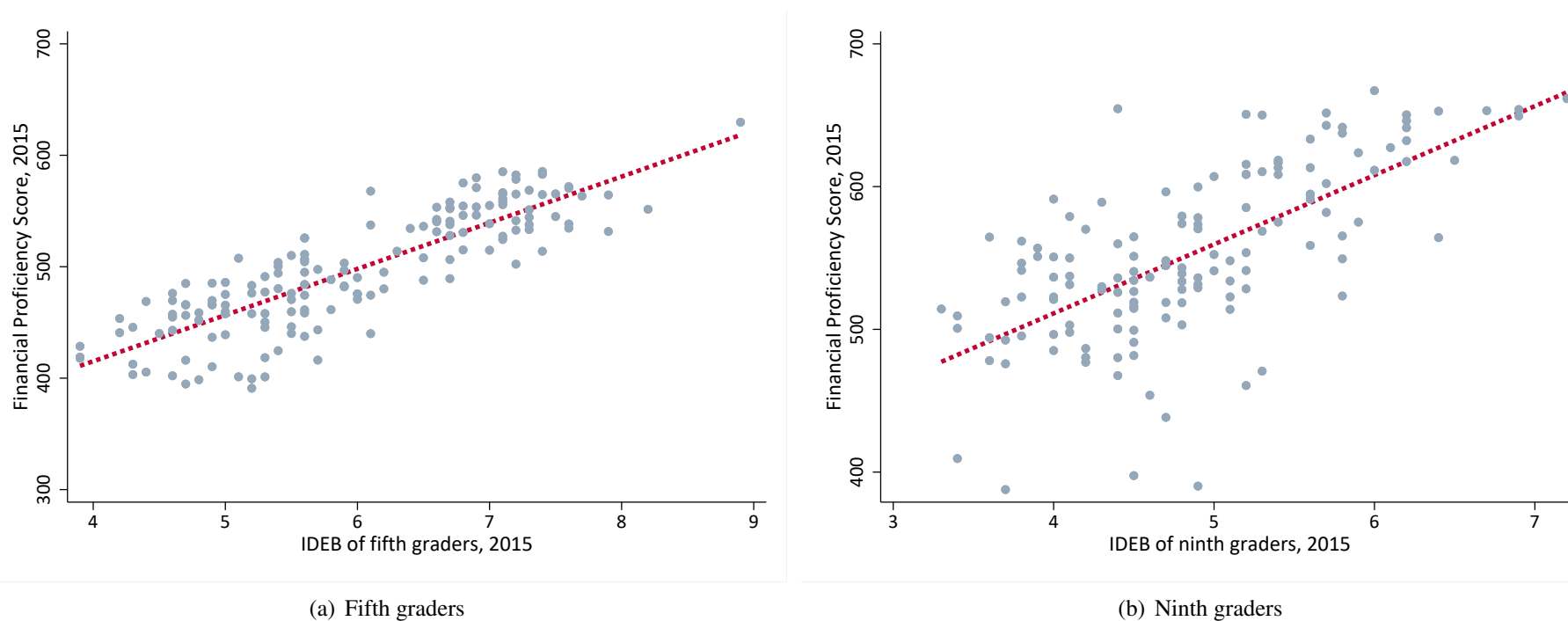
	7th grade				9th grade			
	Strata FE	Strata FE + gender	Strata FE + color of the skin	Strata FE + performance 5th grade	Strata FE	Strata FE + gender	Strata FE + color of the skin	Strata FE + performance 5th grade
Proficiency								
Treatment	0.090 (0.048)	0.090 (0.047)	0.092 (0.048)	0.098 (0.042)	0.107 (0.056)	0.102 (0.055)	0.106 (0.055)	0.149 (0.055)
pvalue	0.061	0.057	0.055	0.022	0.057	0.066	0.056	0.008
Obs.	3640	3637	3622	2538	3374	3362	3346	2251
R2	0.208	0.224	0.209	0.505	0.168	0.183	0.170	0.399
Consumption index								
Treatment	0.080 (0.041)	0.080 (0.040)	0.078 (0.041)	0.070 (0.042)	0.084 (0.037)	0.083 (0.037)	0.084 (0.037)	0.087 (0.044)
RI pvalue	0.054	0.046	0.060	0.098	0.025	0.027	0.025	0.050
Obs.	3409	3407	3394	2384	3163	3160	3149	2115
R2	0.1	0.119	0.1	0.187	0.055	0.061	0.054	0.128
Saving index								
Treatment	0.066 (0.040)	0.064 (0.040)	0.070 (0.040)	0.033 (0.044)	0.018 (0.038)	0.017 (0.038)	0.017 (0.038)	0.030 (0.045)
RI pvalue	0.102	0.112	0.081	0.456	0.644	0.660	0.663	0.508
Obs.	3368	3365	3352	2351	3,105	3,102	3,093	2080
R2	0.032	0.032	0.032	0.065	0.019	0.020	0.020	0.044

Note: $p < 0.01$, $p < 0.05$, $p < 0.1$. Clustered standard errors at the school level are in parentheses. RI stands for Randomized Inference.

The financial proficiency is computed by CAEd based on student's answers on a standardized exam that uses the Item Response Theory (IRT). The consumption and saving indices are based on students' choices in questions containing four possible answers (I totally agree, I agree, I disagree, I totally disagree). The answers are classified on a 1 to 4 scale, where 4 represents the most forward-looking financial behavior. The financial proficiency, savings, and consumption indices are normalized so that the treatment effect can be measured in terms of standard deviations (SD).

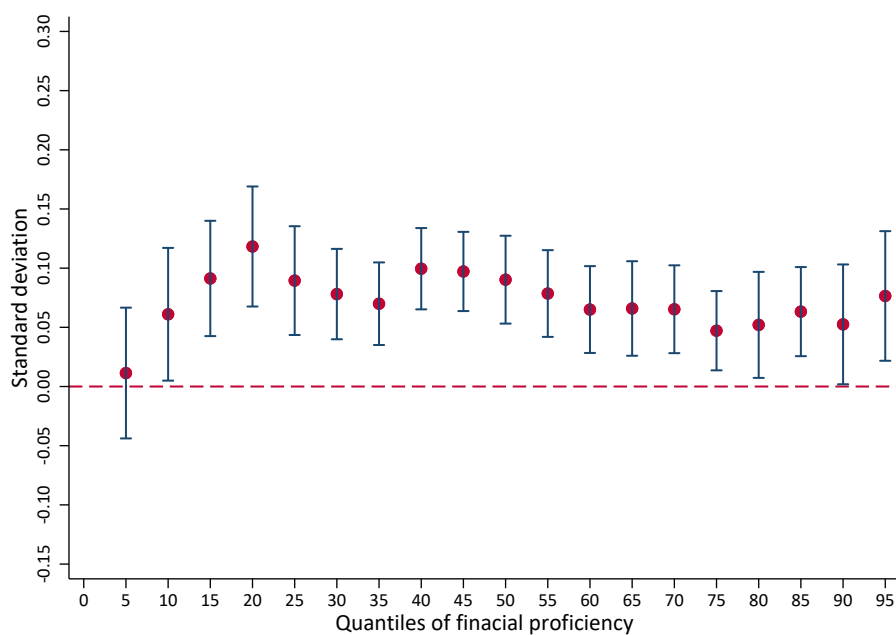
B Figures

Figure B.1: Financial Proficiency Score and IDEB

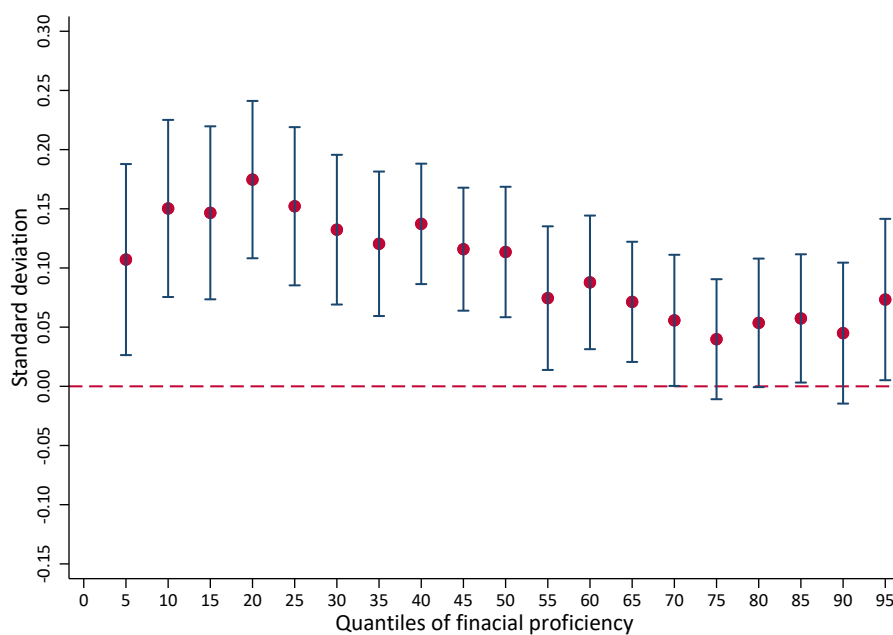


Note: The Figure shows on the y-axis the financial proficiency score calculated by CAEd for the sample of pilot schools and on the x-axis the Education Development Index (IDEB) calculated by the Brazilian National Institute of Education and Research. IDEB is the most important educational indicator in Brazil. The index is calculated by multiplying students' proficiency in Portuguese and Math (on a scale of 0 to 10) by grade promotion (on a scale of 0 to 1).

Figure B.2: Quantile ITT estimates for financial proficiency



(a) Pooled, elementary and middle school



(b) Middle school

Source: The financial proficiency is computed by CAEd based on student's answers on a standardized exam that used the Item Response Theory (IRT). The index is normalized so that the treatment effect can be measured in terms of standard deviations (SD). We first run this financial proficiency score on the treatment dummy, the six strata dummies, and on a constant. We then take the residual of this estimation and run a simultaneous quantile regression, for each one of the quantiles shown in the Figure, in which the dependent variable is the residual of the first regression and the independent variable is the treatment dummy. 1000 bootstrap replications.